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Real-Time Artifact Recognition during ECG Acquisition

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Tag der Anmeldung: 5. January 2024
Tag der Abgabe: 21. March 2024

Ich erkläre hiermit, dass ich die vorliegende Arbeit selbständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel verwendet habe.

Bremen, den 21. March 2024

Zusammenfassung

Die Qualität des Electrokardiogram (EKG) ist entscheidend für medizinische Diagnosen, wissenschaftliche Forschung und Gesundheitsüberwachung. Die Erkennung und Minimierung von Artefakten und Rauschen während der EKG-Signalerfassung stellt daher eine fortwährende Herausforderung dar.

Diese Studie untersucht Artefakte und Störungen, die durch menschliches Versagen oder fehlerhafte Handhabung während der EKG-Erfassung entstehen, wie z.B. Elektrodenablösung, unsachgemäße Verwendung von Elektroden, unerwartete magnetische Feldstörungen und Bewegungen der Probanden. Diese Störungen können unbrauchbare und irreparable Signale erzeugen, die oft von den Erfassern unbemerkt bleiben. Sie können jedoch durch Modifikation der Erfassungsverfahren und Umgebungsbedingungen gemildert oder beseitigt werden. Die Erkennung von Artefakten während der EKG-Erfassung trägt maßgeblich zur Lösung dieses Problems bei. Nach Identifizierung eines bestimmten Artefakt- oder Störungstyps können die Experimentatoren sofort entsprechende Maßnahmen ergreifen, um das Problem effizient zu lösen.

Aufbauend auf diesem Konzept schlägt die Studie die Echtzeit-Erkennung von Erdungsartefakten während der EKG-Erfassung vor. Verschiedene Arten von Erdungsartefakten und Rauschen dienen als Forschungsschwerpunkt, wobei für jede Art von Artefakten Simulationsmethoden entwickelt werden. Ein Datensatz von zehn Probanden wird gesammelt und geteilt, Signale werden anhand simulierter Ereignisse kalibriert, und maschinelles Lernen wird erkundet. Geeignete Modelle des maschinellen Lernens werden für das Training ausgewählt und auf Echtzeit-Erkennungsaufgaben angewendet.

Verschiedene Erkennungsarchitekturen, einschließlich Einzel-, Zwei- und Mehrschichtmodelle, werden vorgeschlagen, um die Leistung des Modells zu verbessern. Durch Zuweisen von Prioritätserkennungsstufen für verschiedene Artefaktkategorien wird die Leistung des Modells verbessert. Insbesondere in dem Experiment, das die bevorzugte STD-Binärklassifikation verwendet, übertrifft die Erkennungsgenauigkeit von STD in der Einzelschichtarchitektur.

Das Offline-Modell erreicht eine Erkennungsrate von 90,89% in der 5-besten Ausgabe-Kreuzvalidierung unabhängig von menschlicher Intervention. Durch Extrahieren bestimmter Klassen wie TA-M und TA-S wird die Erkennungsrate des Offline-Modells auf 96% in der 5-besten Ausgabe-Kreuzvalidierung unabhängig von menschlicher Intervention verbessert.

Abstract

The quality of electrocardiogram (ECG) signals is pivotal for medical diagnosis, scientific research, and health monitoring. Detecting and mitigating artifacts and noise during ECG signal acquisition pose ongoing challenges.

This study investigates artifacts and noise arising from human errors or mishandling during ECG collection, such as electrode detachment, improper electrode usage, unexpected magnetic field interference, and subject movements. These disturbances can generate unusable and irreparable signals, often escaping detection by collectors. However, they can be mitigated or eliminated by modifying collection procedures and environmental conditions. Recognizing artifacts during ECG acquisition greatly aids in addressing this issue. Upon identifying a specific type of artifact or noise, experimenters can promptly implement corresponding measures to efficiently resolve the problem.

Building upon this notion, the study proposes real-time recognition of grounding artifacts during ECG acquisition. Various types of grounding artifacts and noise serve as the research focus, with simulation methods devised for each type. A dataset from ten subjects is collected and shared, signals are calibrated based on simulated events, and machine learning solutions are explored. Suitable machine learning models are chosen for training and applied to real-time recognition tasks.

Different recognition architectures, including single-layer, two-layer, and multi-layer models, are proposed to enhance model performance. By assigning priority recognition levels to different artifact categories, the model's performance is improved. Notably, in the experiment employing preferential STD binary classification, the recognition accuracy of STD surpasses that of in the one-tier architecture.

The offline model achieves a recognition rate of 90.89% in 5-best output cross-validation independent of human intervention. Extracting specific classes such as TA-M and TA-S boosts the offline model's recognition rate to 96.11% in 5-best output cross-validation independent of human intervention.

Contents

1 Introduction **1**

2 Fundamentals **3**
2.1 Biosignals . . . 3
2.2 ECG . . . 4
2.3 Artifacts in ECG . . . 6
2.4 Application in real-time . . . 7
2.5 Machine learning . . . 7
2.5.1 Decision tree . . . 9
2.5.2 Support vector machines . . . 9
2.5.3 Random forest . . . 9
2.6 Leave-one-out cross validation . . . 10

3 State-of-the-art **11**

4 Experimental devices and data collection **13**
4.1 Experimental devices . . . 13
4.2 Data acquisition . . . 14
4.2.1 Protocols design . . . 14
4.2.2 Precautions during data acquisition . . . 15
4.3 Dataset description . . . 18
4.3.1 Sensor placement . . . 18
4.3.2 Subjects . . . 21
4.3.3 Ethic, privacy preservation and data security . . . 21
4.3.4 Data format . . . 21
4.4 Data annotation . . . 22
4.4.1 Events-based . . . 22
4.4.2 Pattern-based . . . 23

5 Data Preparation **25**
5.1 Data segmentation with sliding windows . . . 25
5.2 Significance of features extraction for machine learning . . . 26

6 Experiments, results, and discussions **29**
6.1 Person-dependent experiments . . . 31
6.1.1 Window length and overlap ratio selection . . . 33
6.1.2 Feature selection . . . 34
6.1.3 Classification results . . . 35
6.1.4 Discussion . . . 38

6.2 Semi-Person-Independent experiments 39
6.2.1 Classification results 39
6.2.2 Discussion 39
6.3 Person-Independent experiments 41
6.3.1 Classification results and discussion 41
One-tier and hierarchical modeling 41
N-Best outputs 44
6.3.2 Results and discussion 44
6.4 From offline model towards real-time recognition 48
6.4.1 Discussion 48
6.5 Pattern-based classification 52
6.5.1 Annotation 52
6.5.2 Feature selection and training 53
6.5.3 Results and discussion 53

7 Conclusion **57**

8 Future work **59**

A Acronyms **61**

B Appendix **63**
B.1 The 10-Subject dataset 64
B.2 Experiments 65
B.2.1 Person-dependent experiment results 65
B.2.2 Semi-person-independent experiments with 11 statistical features, LOOCV, without BAR 68
B.2.3 Person-independent experiments with 11 statistical features, LOOCV, without BAR 71

Bibliography **75**

List of Figures

2.1 Electrocardiogram (ECG) waveform 5

4.1 ECG sensor placement 19

6.1 One-tier and hierarchical modeling 30
6.2 Data distribution: small-scale dataset of person-dependent (PD) and semi-person-independent (semi-PI) experiments 31
6.3 Confusion matrix of the one-tier support vector machine (SVM) with the best accuracy in person-dependent experiments 36
6.4 Heatmap of combinations of window length and overlap ratio of the one-tier support vector machine (SVM) in person-dependent experiments 37
6.5 Example of masking electrodes to simulate the gel deficiency situation 39
6.6 Data distribution of 10-subject dataset 41
6.7 Comparison results of with and without 50% STD in one-tier modeling 42
6.8 Comparison results of recognition accuracy with one-tier architecture and with two or hybrid multiple layers architecture 43
6.9 Recognition rates of one-tier person-independent N-best leave-one-out cross-validation (LOOCV) experiments 44
6.10 Confusion matrix of the one-tier support vector machine (SVM) . . . 45
6.11 Confusion matrix of the support vector machine (SVM) modeling excluding TA-M and TA-S 46
6.12 Screenshot of a running demonstration for real-time artifact detection and classification during ECG acquisition 49
6.13 Instance of the joint window length and overlap ratio selection using heatmap 50
6.14 Data distribution: pattern-based annotation, 10-subject dataset . . . 54
6.15 Confusion matrix of the one-tier random forest (RF) modeling with all classes 55
6.16 Confusion matrix of the one-tier random forest (RF) modeling with classes without STD and corrupted ECG 56

List of Tables

4.1 Taxonomy of real-time artifacts during ECG acquisition 16
4.2 ECG sensor electrodes position of various events 20

5.1 All 30 used features in each domain from TSFEL library[RLF+22] . . 27

6.1 Taxonomy of real-time artifacts during ECG acquisition: old abbreviation 32
6.2 The most optimal combination of window length and overlap ratio for decision tree (DT), support vector machine (SVM) and random forest(RF) in person-dependent (PD) experiments 33
6.3 Classification accuracy of decision tree (DT) and support vector machine (SVM) across statistical, temporal and frequency feature domains 34
6.4 Person-dependent (PD) experiments: Best accuracy of decison tree (DT), support vector machine (SVM), random forest (RF) with parameters list 36
6.5 Person-dependent experiments: accuracy of STD recognition with dataset in three handling of BA-R class 36
6.6 Recognition rates of semi-person-independent(semi-PI) experiments with the random forest (RF) and the support vector machine (SVM) modeling for comparing the performance of one-tier pure person-dependent (PD) with 4 folds and semi-person-independent (semi-PI) with 7 folds training with 11 statistical domain features. semi-PI values inferior to PD are indicated in gray. 40
6.7 Recognition rates of N-best leave-one-out cross-validation experiments with the random forest (RF) and the support vector machine (SVM) modeling for comparing the performance of one-tier pure person-independent (PI) and semi-person-independent (semi-PI) training with 11 statistical domain features 47
6.8 Taxonomy of real-time artifacts during ECG acquisition: pattern-based 52
6.9 Recognition rates of N-best leave-one-out cross-validation and binary classification between standard ECGs and corrupted ECGs experiments with the random forest (RF) modeling for comparing the performance of one-tier pure person-independent (PI) on event-based labels and pattern-based labels with 11 statistical domain features . . 55

B.1 Details of 10-subjects dataset 64
B.2 Recognition results of person-dependent experiments 66
B.3 Recognition results of person-dependent experiments 67
B.4 Binary and triple recognition results of person-dependent experiments 69
B.5 Binary and triple recognition results of person-dependent experiments 70

B.6 Recognition results of semi-person-independent experiments 71
B.7 Recognition results of person-independent experiments 72
B.8 Recognition results of person-independent experiments 73

1. Introduction

Biosignal acquisition marks the inception of all further investigations into biosignal analysis. The quality of this starting point determines the complexity of subsequent research endeavors and the ultimate success of the investigation. Notably, biosignals are highly susceptible to various artifacts and noise, making noise filtering a focal point of numerous research endeavors. However, certain types of noise, such as low-frequency noise from muscle activity, pose challenges to filtering without compromising the integrity of the original biosignals' diagnostic information. Therefore, during acquisition, all potential sources of disturbance affecting the biosignal's quality must be promptly addressed to mitigate irreparable damage before storing the data in the dataset.

While some artifacts or noises exhibit observable patterns that can be identified by signal observation and expert listening, many others are challenging to discern with the naked eye. Moreover, with the widespread use of biosignals in daily life by non-experts, there is an increasing demand for automatic artifact recognition. In response to this need to ensure high-quality data, this study proposes a novel approach that considers various types of artifacts and noise as research objects.

ECG signals are among the most well-known biomedical signals, having been studied for over a century[OSÓ22]. Over the past two decades, there has been a significant surge of interest in developing efficient methods for processing and analyzing ECG signals to extract useful diagnostic information. ECG signals are quasi-periodic, with relatively low amplitudes of several millivolts, and are often affected by noise [Gac12]. However, due to their repeated characteristic pattern and single-channel nature, the variations in their noise and artifacts are relatively limited, making the simulation of ECG noise and artifacts feasible. Thus, ECG serves as an ideal biosignal for validating the novel approach proposed in this study, simplifying the complexity associated with biosignals.

In this study, the primary sources of interference causing ECG artifacts were simulated as events, including electrode detachment, insufficient gel on electrodes, unanticipated magnetic field interference, and signal distortion due to human movements. Often unnoticed during acquisition, machine learning algorithms can recognize these artifacts in real-time. Additionally, a classification of noise based on distinctive patterns was implemented, although it is worth noting that similar patterns may correspond to different events, posing challenges for machine learning models.

The research framework utilized in this study offers valuable insights for investigating other types of biosignals and can be extended to electrodermal activity (EDA), respiratory inductance plethysmography (RIP), etc. The study provides a taxonomy of ECG acquisition artifacts and simulation schemes and a dataset comprising ten subjects for further research on ECG noise and artifacts. Multiple lightweight machine learning models, layered architectures, and parameter configurations were explored to achieve offline detection and classification, with the method's real-time applicability successfully validated. This study focuses primarily on artifacts, encompassing a few noise types, such as electromyography (EMG) noise and high-frequency noise. Throughout this paper, the subjects will be referred to as artifacts, clarifying this distinction when discussing specific types of noise.

2. Fundamentals

2.1 Biosignals

Biosignals, as autonomous and physically measurable quantities produced by living organisms, serve as objective descriptors of physiological states [SM23]. In Germany, biosignals are officially defined as "Information that emanates from physical (possibly chemical) actions of the human body"[1]. In fact, biosignals in the academic sense include bothelectrical and non-electrical signals [Liu21]. They are widely employed in disease diagnosis, biomedical research, and human-computer interaction, facilitating the monitoring and comprehending human physiological health states. These signals are typically collected using specialized sensors, establishing the groundwork for subsequent analysis. However, during the process of signal acquisition, various physical factors like environmental conditions and equipment, along with biological factors from the human body, may introduce artifacts into the biological signals. These additional elements could interfere with signal analysis, complicating the interpretation of the data.

The quality of biosignals is therefore paramount for ensuing research, yet it poses a challenge for all collection efforts. High-quality signals effectively convey the biological information they encapsulate, whereas lower-quality signals can lead to uncertainty and interference during subsequent analyses. Although human observation can empirically identify some low-quality signals, the reliability of such observation remains questionable.

[1]Translated from the definition of DIN 44300, German Institute for Standardization.

2.2 ECG

An electrocardiogram (ECG; In German, the electrokardiogram, EKG) is a measure of how the electrical activity of the heart changes over time as action potentials propagate throughout the heart during each cardiac cycle [DVI05]. ECG is a non-invasive diagnostic modality that has a substantial clinical impact on investigating the severity of cardiovascular diseases [RCOB18]. The continuous monitoring of dynamic changes in physiological and biological parameters through non-invasive integrated systems can serve a crucial role across a diverse range of applications, encompassing healthcare and sports training fields [SSB+15][PCC+15][OGP+15]. Precise and prompt acquisition and interpretation of ECGs enhance cardiac monitoring quality, minimizing the risk of misdiagnosis and treatment delays, particularly critical in cardiovascular disease, a leading cause of mortality.

ECG acquisition involves placing electrodes on the skin to record the heart's electrical activity over some time. A typical ECG waveform includes two intervals: (a) PR interval including both the P waveform and PR segment; (b) the QT interval including Q, R, and S waveforms (QRS complex), ST segment, and T waveform, respectively [SB11]. Figure 2.1[2] illustrates a typical ECG waveform.

ECG conveys information regarding the heart's electrical function by altering the shape of its constituent waves, namely the P, QRS, and T waves. Thus, the required tasks of ECG processing are the reliable recognition of these waves and the accurate measurement of clinically essential parameters measured from the temporal distribution of the ECG constituent waves [MSD+98]. In this study, ensuring that the collected ECG preserves essential parameters for data analysis or clinical studies, such as the quality of the fundamental waveform, emerges as a critical criterion for evaluating the signal quality of the ECG.

[2] Basic ECG waveforms, intervals and segments from https://www.wikilectures.eu/w/Electrocardiogram

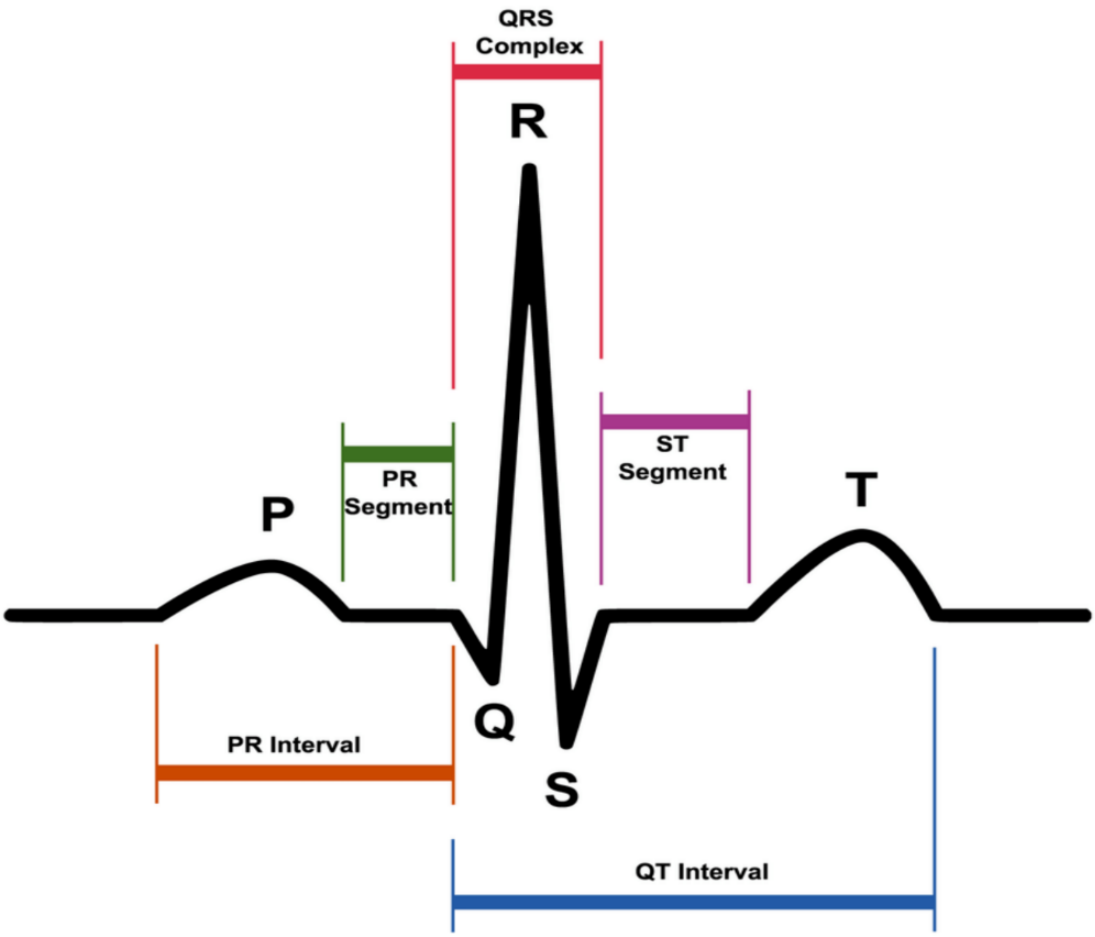

Figure 2.1: A typical electrocardiogram (ECG) waveform and its characteristic patterns (P and T waves, PR and ST segments, PR and QT intervals, as well as the QRS complex)[Mmfn09].

2.3 Artifacts in ECG

The ECG signal sampling process is often influenced by various types of signal corruption, including artifacts stemming from the patient's body movements or micro-movements, electromagnetic interference, sensor malfunctions, and issues related to electrode effectiveness. Unfortunately, artifacts in ECG can be within the frequency band of interest and manifest with similar morphologies as the ECG itself [Cli06]. The ECG artifacts mainly involved in this study can be classified as follows[Cli06]:

- Power line interference: 60Hz
- Insufficient gel on electrode: Insufficient gel causes loss of contact between the electrodes and skin
- Electrode motion artifacts: The movement of the electrode away from its contact area on the skin can result in impedance variations between the electrode and the skin, potentially causing fluctuations in the ECG signal. This typically occurs when the body part connected to the electrode pad undergoes movement.
- EMG Noise: Electrical activity caused by muscle contractions lasting around 50 ms
- Baseline drift: Usually caused by deep respiration or chest/body movement
- High-frequency noise: Noise could be generated by other equipment. It lasts short and happens irregularly
- Electromagnetic interference (EMI): Interference occurs when the subject/patient remains close to the power supply during acquisition, typically persisting throughout the process.

Hence, artifacts can affect subsequent signal analysis to different extents, highlighting the importance of identifying and controlling them during the signal acquisition.

2.4 Application in real-time

The offline model can identify artifacts while concurrently enabling the reduction of noise weight during the preprocessing, often accomplished through filters. However, implementing this process increases the workload before signal analysis, and in some cases, signals may require resampling due to substantial artifacts. Consequently, this also extends the time investment to a certain degree.

Incorporating a real-time model provides a rapid response mechanism whereby a machine learning algorithm assesses signal quality during data collection, making real-time adjustments to mitigate artifacts. Should artifact levels reach a critical threshold, potentially affecting key analytical parameters like the P-wave, QRS complex, and T-wave, the real-time model prompts immediate data recollection. This approach streamlines decision-making, swiftly identifies issues, offers guidance, and proposes solutions, ensuring high-quality data acquisition without extensive preprocessing.

A strictly defined real-time recognition program must guarantee a response within specified time con-straints, often referred to as "deadlines"[BA06]. It controls an environment by receiving data,processing them, and returning the results sufficientlyquickly to affect the environment at that time[Mar65]. In the perspective of digital signal processing (DSP) of real-time recognition systems,the analyzed (input) and generated (output) samples should be processed (or generated) continuously in the time it takes to input and output the same set of samples independent of the processing delay[KLT17]. [Liu21] used a simple quantitative example to explain this issue: if an HAR system requires more than 1 second to process, recognize, and respond to a 1-second window/frame of recorded signals, the user will feel the system's recognition outputs slower and slower. Thus, after a short duration, it becomes a "congested" system that disrupts the real-time experience. If window overlapping, an often applied technology in artifacts recognition for improving recognition accuracy, is taken into account, the processing time for each input required for a real-time recognition system is even more demanding.

In the real-time model developed in this study, the trained machine learning model conducts artifact type recognition in real-time during the ECG signal collection process, utilizing the live display of the signal. While signal visualization provides real-time monitoring of signal acquisition, it also synchronously and promptly informs the user about the identified type of workpieces and suggests corresponding solutions. This allows the user to decide whether to pause the collection or make adjustments based on the provided guidance.

2.5 Machine learning

The core of machine learning is the automatic discovery and use of patterns in data. These patterns can be used to cluster or predict classes or data points. The classification of classes is called classification, and the prediction of logarithmic points is called regression.

This paper mainly studies supervised classification, that is, the classification of artifacts into pre-defined classes. In supervised classification, samples must be labeled in advance; different artifacts correspond to the features of different sample

patterns. The classifier learns the patterns in the training set samples and then classifies them according to the features of the samples in the test set. Therefore, in data preprocessing, the selection and calculation of sample features can improve the learning efficiency and effect of the classifier. Machine learning results can validate the method proposed in this study. It also provides a comparable baseline for subsequent deep-learning results.

As a proof-of-concept, all recent experiments explored three no deep ML models, decision tree (DT), support vector machine (SVM), and random forest (RF), using window-based data training and recognition.

2.5.1 Decision tree

The decision tree (DT) algorithm is renowned for aligning with human intuitive thinking and proficiency in solving classification problems. Its interpretability and straightforward implementation contribute to its wide-ranging applicability. DT algorithm delineates classification paths by operating via a hierarchical if-then-else approach on feature attributes. Comprising root nodes, internal nodes, and leaf nodes, the root node encapsulates the entire sample set. In contrast, internal nodes autonomously represent attribute-based judgments derived from training data, obviating the need for manual configuration. Leaf nodes encapsulate classification outcomes.

DT-based algorithms offer several advantages, including the ability to process irrelevant features, swift execution, and the prompt generation of practical results, even with vast datasets. DTs are particularly useful as an initial model for early artifact classification, adeptly handling numerous features without requiring refined feature selection. Moreover, their computational efficiency renders them well-suited for real-time analysis, thereby justifying their utilization as a training model in this study.

2.5.2 Support vector machines

The primary objective of support vector machines (SVM) is to identify a hyperplane that maximizes the separation between data points and potential classes within an n-dimensional space. The data point closest to this hyperplane, known as the support vector, plays a crucial role in defining the decision boundary [Shm05].

While SVM excels in binary classification tasks, it doesn't inherently support multi-class classification. To address this, a common approach involves decomposing the multi-class problem into several binary classification problems.

One method for achieving this is the one-to-one method, where data points are mapped into a high-dimensional space to achieve linear separation between every pair of classes. Alternatively, the One-to-Rest method assigns one binary classifier per class.

Despite its advantages in handling high-dimensional data, SVMs are sensitive to noise and outliers, increasing the risk of overfitting. Additionally, SVM models often require extensive computational resources, limiting their suitability for real-time analysis. These considerations are crucial for optimizing the implemented model effectively.

2.5.3 Random forest

Random forests (RF) assemble collections of DTs to yield more accurate and stable predictions. They are typically trained using the bagging method, which enhances classification outcomes by learning from a combination of models. In contrast to traditional decision trees, RF prioritizes identifying the best features within randomly selected subsets, fostering a broader diversity that often results in more effective models. Unlike DTs, the RF algorithm constructs multiple DTs by randomly selecting observations and features and averaging the results.

Additionally, the RF algorithm addresses the overfitting deep DTs encounter when faced with excessive feature attributes. This is achieved by creating random feature subsets to construct smaller trees, mitigating overfitting. However, this operation may result in slower computations. Hence, achieving a balance between computational speed and classification efficacy is crucial for real-time applications to obtain the most suitable model for analysis.

2.6 Leave-one-out cross validation

In pursuit of a model with broad utility, the transition from a person-dependent to a person-independent model occurs once the former achieves satisfactory results. All experiments employ leave-one-out cross-validation (LOOCV) to validate the person-independent models comprehensively. The overall recognition rate is determined as a macro average by aggregating ten-fold results.

The dataset comprises data from 10 participants. During each training iteration, data from 9 out of the ten individuals are utilized for training, while the data from the remaining individual is reserved for testing. This approach ensures that no participant-related data overlaps between the training and test sets, minimizing inherent bias in the test set predictions. The model's average accuracy is then calculated to gauge its performance.

To account for the biological variability among individuals, cross-over experiments were conducted to ensure the reliability and impartiality of the results. This methodology enables the derivation of the model's average performance. Despite the relatively modest size of the dataset, this approach provides a fair evaluation of the model's capabilities. Nevertheless, as the dataset size expands, this method has significant computational costs, which is worth considering in future works.

3. State-of-the-art

Dealing with artifacts and noise during ECG acquisition and processing has posed a challenge. In current practice, two primary approaches have been proposed to address the issue of ECG noise and artifacts. One approach involves using post-acquisition tools, techniques, and algorithms for detecting, reducing, or eliminating ECG artifacts and noise, and assessing the acceptability of the recorded ECG [SRM18]. Various methods are employed to achieve detecting, reducing, or eliminating ECG artifacts and noise, including the implementation of frequency and time domain filters. Wiener filtering and Kalman filtering methods are the pioneering works that aim to remove the additive noises. However, due to the non-stationary nature of the ECG signal, frequency domain filters could cause distortion in a transient interval of the signal and important clinical information may be lost [AD08] [LBH+09]. Denoising methods for ECG signals based on wavelet techniques [Don95][ASMK09][CK01][SS06] have also been continuously proposed. Ling et al. suggested a fuzzy rule based multiwavelet ECG signal denoising [WKLYFHL+08]. A method which uses an ECG signal denoising by wavelet transform thresholding was presented by Alfaouri and Daqrouq [AD08]. Sharma et al. [SDM10] developed an ECG signal denoising using higher order statistics in wavelet subbands.

Assessing the acceptability of the signal during ECG signal acquisition enables prompt identification and resolution of issues as they arise. It saves time and labor costs by avoiding the acquisition of unusable signals while ensuring signal quality. Commonly used methods can use automated algorithms to detect artifacts and quantify the quality of the recorded signal [IMSR16]. Detection methods include simple binary categorization as clean or contaminated. One example is at the front end of the acquisition process. The user is provided with quick binary feedback and can adjust the recording settings if needed, or in the worst case restart the recording [MSM+19]. Lee presented a real-time method for the detection of motion and noise artifacts, which discriminated between clean and noise-contaminated ECG segments [LMMC12]. Kumar and Sharma presented to detect and classify different electrocardiogram (ECG) noises including baseline wander (BW), muscle artefact (MA), power line interference (PLI) and additive white Gaussian noise (AWGN) based on signal decomposition on mixed codebooks [KS20]. The 2011 Physionet/Computing in Cardiology (CinC) Challenge [XGM+11], [ZHM11], [Moo11] aimed at encouraging

the development of cell phone software by recording ECGs and providing useful feedback about their quality [SMC11]. The best-performing algorithms consist of several stages, namely: flatline detection, missing channel identification, and automatic and cross-correlation thresholding [XGM+11].

According to different research objectives and different application requirements for signals, multilevel quantization of signal quality has been proposed [RXC+12], [VIGB10]. For example, Li et al. proposed a five-level signal quality classification algorithm to categorize signals into five classes: clean, slight noise, moderate noise, severe noise, and extreme noise [LRC14]. Moeyersons et al. proposed a ECG quality assessment based on the posterior probability of the clean class as an indication of the quality of the signal [MSM+19].

While these techniques can address some artifacts and noise, extreme noise might lead to the loss of ECG information or even the destruction of the target segment. In such cases, signal quality can only be assured through reacquisition. Although the online binary clean ECG recognition method can save time and labor costs, allowing users to decide whether to reacquire noisy ECGs during acquisition, it does not provide guidance on adjusting existing acquisition methods for better signal quality.

This study proposes a novel approach by focusing on different types of artifacts and noise rather than clean ECGs. By simulating various events that cause artifacts and noise, a dataset is created. Machine learning is then employed to train a model capable of identifying artifacts and noise in real-time during ECG collection, offering appropriate solutions to users. This approach provides an effective solution for handling artifacts and noise during real-time collection, saving labor and event costs while enhancing signal quality compared to post-acquisition processing. Furthermore, the present method, in addition to binary classification and narrow-range pattern-based recognition for noise and artifacts, utilizes event-triggered classification, which encompasses a broader spectrum of noise and artifact types with greater classification detail. This approach offers users more intuitive solutions.

4. Experimental devices and data collection

4.1 Experimental devices

In the present investigation, the biosignalsplux Researcher Kit [1] served as the designated biosignal recording apparatus, leveraging its sophisticated design to provide versatile solutions through the integration of hot-swappable sensors and automatic synchronization. A cornerstone of this instrumentation, the biosignalsplux hub, demonstrated its pivotal role by facilitating concurrent signal acquisition across eight channels, thereby accommodating the integration of sensors chosen at discretion. This exceptional feature empowered the seamless integration of a diverse array of arbitrarily selected sensors, elevating the versatility of our data collection methodology. The feasibility and the stability of the biosignalsplux hubs and their attached sensors during acquisition has been validated by several preliminary research, such as accelerometers (ACC), electromyography sensor (EMG), electrogoniometer (GYRO), piezo and airborne microphone in various perspektive of human activity recognition[HLS23] [RAGS13][Pal15][HLS20], electroencephalogram sensor (EEG), functional near-infrared spectroscopy sensor (fNIRS), electrocardiogram(ECG) sensor, electrodermal activity sensor (EDA) and respiratory inductance plethysmography (RIP) in attention classification during standard cognitive tasks for occupational domains[GVR+22], finger photoplethysmography sensor (PPG) on reconstitution of continuous arterial blood pressure [SZZ+23] and electromyography sensor (EMG) in facial action unit recognition[VLK+24] etc.

In the study conducted, as part of the investigation, a single channel of the biosignalsplux hub, among the eight available channels, was designated for capturing electrocardiogram (ECG) signals. The ECG signals were acquired using the Biosignalsplux ECG sensor [2], configured with low-noise ECG local differential triodes. This sensor is specifically designed for extracting heart rate data, analyzing nuances within the data, and providing medical-grade raw sensor data. The ECG signals

[1] https://www.pluxbiosignals.com/products/researcher-kit (accessed 26 Dec, 2023)

[2] https://www.pluxbiosignals.com/products/electrocardiography-ecg-sensor-1 (accessed 26 Dec 2023)

were acquired in this study at a high sampling rate of 1000 Hz, ensuring temporal precision and with a quantization level of 16 bits, providing a satisfactory resolution of signal amplitudes.

The unutilized seven-channel interfaces within the biosignalsplux hub in the current study harbor significant potential for future research endeavors. These unused channels provide an opportunity to delve into multichannel and multimodal biosignal acquisition, artifact detection, and exploring synergy or antagonism between signals. This forward-thinking approach is pivotal for advancing the scope of biosignal research, enabling the integration of diverse data streams and facilitating the exploration of complex biosignaling interpretations of physiological phenomena.

It is crucial to recognize that the specific model and manufacturer of the biosignalsplux Researcher Kit and the utilized biosignalsplux ECG sensor may introduce nuances that could impact the generalizability of the study's findings. However, by adhering to the proposed taxonomy, acquisition processes, and experimental procedures outlined in this study, the results are conducive to replicating alternative equipment models. Following the established methodology allows researchers to anticipate similar outcomes on different biosignal recording devices, thereby contributing to the robustness and broader applicability of the findings.

4.2 Data acquisition

To emulate diverse types of artifacts during data acquisition, the data acquisition protocol has been meticulously designed to generate artifacts while upholding a stringent, accurate, and reproducible experimental methodology. Factors contributing to artifact appearance assessed comprehensively are categorized into two primary groups: physiological artifacts (BA-) and technical artifacts (TA-). Physiological factors encompass artifacts induced by the frequency and amplitude of the participant's breath during signal acquisition. Additionally, the database incorporates high-quality ECG signals devoid of artifacts.

Considerations in the signal acquisition process merit careful attention as they can significantly impact both artifact and average signal quality. For instance, introducing other artifact types during a single artifact simulation or causing inaccuracies that interfere with subsequent identification are potential consequences of overlooking considerations in the signal acquisition process. This nuanced approach to artifact simulation and signal acquisition ensures the reliability and validity of the study's outcomes.

4.2.1 Protocols design

The artifact-inducing factors are categorized into two main groups: biological artifacts (BA-) and technical artifacts (TA-). And standards ECG signals without artifacts (STD) are also included in the database. Each class of TA-containing signals from each subject was acquired continuously for 30 seconds, while each BA was captured in two sessions of 15 seconds each, adhering to limitations imposed by human physiology. Additionally, STDs were acquired twice from each subject, with each session lasting 60 seconds, once in a seated position and once in a standing position.

Biological factors include artifacts induced by the frequency and amplitude of the participant's breath during signal acquisition. Three different types of breath are

defined in protocol, which are holding breath (BA-R-H), fast breath (BA-R-F), and deep breath (BA-R-D).

Technical factors include movement of the device (TA-S), dropping of the electrodes (TA-D), exchange of left and right electrodes (TA-C-S), overuse of the electrodes (TA-E-O), misplacement of the electrodes (TA-M), interference from electromagnetic fields (TA-I) and insufficient gel of the electrodes (TA-E-IG).

TA-S pertains to artifacts resulting from the shifting of electrodes due to human motion during signal acquisition. For instance, participants engaged in activities such as chest expansion (TA-S-C), hand raising (TA-S-A), squatting (TA-S-S), forward body bending over (TA-S-B), jumping, and waist lateral twisting (TA-S-W) during ECG signal acquisition, inducing electrode displacement.

TA-E denotes electrodes detachment during data acquisition. TA-C-S refers to switched positive and negative cables. TA-E-O refers to overused positive, negative, and/or grounding electrodes, possibly covered by dust. TA-M signifies electrodes placed on active muscles, such as quadriceps and Sternocleidomastoid. Incorrect placement on muscles can result in the mixing of the EMG signal with the ECG signal. Thus, accurate positioning on the clavicle for positive and negative electrodes and on the anterior superior iliac spine for the grounding electrode is essential.

TA-I stems from interference caused by electromagnetic fields, such as power lines, computer battery relays, and cell phone charging cables. Insufficient gel is simulated by partially covering electrodes with plastic stickers (1/8, 1/4, 1/2, 3/4, 7/8), intentionally limiting gel contact.

The detailed taxonomy, as listed in Table 4.1, includes the magority of common events which may cause artifacts during biosignal acquisition.

4.2.2 Precautions during data acquisition

Improper manipulation during the acquisition of the ECG signal may degrade the signal quality and complicate the composition of the artifacts, thus affecting the recognition of artifacts. Here are the precautions summarized in the ECG signal acquisition process, which can help improve the data acquisition quality.

Precautions:

- Remove potential sources of electromagnetic interference from the environment, such as cell phones and computers.
- Instruct participants to wear lightweight clothing to facilitate electrode placement and prevent clothing from tugging on the electrodes.
- Clean the area of electrode placement from any hair.
- Before electrode placement, cleanse the skin with alcohol to eliminate oil residue.
- Inquire about any existing heart conditions with participants.

Table 4.1: Taxonomy of real-time artifacts during ECG acquisition. In each abbreviation, the first term indicates the artifact category (technical or biological artifact); the second points out where the problem to be examined lies; the third specifies further details[LZG+24].

TA-*: Technical Artifacts	
Abbreviation	ECG Aquisition Artifact
TA-C-S	**C**able **S**witched: positive and negative
TA-D-P	**D**etachment: **P**ositive electrode
TA-D-N	**D**etachment: **N**egative electrode
TA-D-G	**D**etachment: **G**rounding electrode
TA-D-PN	**D**etachment: **P** and **N** electrodes
TA-D-PG	**D**etachment: **P** and **G** electrodes
TA-D-NG	**D**etachment: **N** and **G** electrodes
TA-E-IG	**E**lectrodes with **I**nsufficient **G**el
TA-E-O	**E**lectrodes **O**verused: **P**, **N**, and **G**
TA-I	Magnetic field **I**nterference
TA-M-P	**P**ostitive electrode on active **M**uscles
TA-M-N	**N**egative electrode on active **M**uscles
TA-M-G	**G**rounding electrode on active **M**uscles
TA-M-PN	**P** and **N** electrodes on active **M**uscles
TA-M-PG	**P** and **G** electrodes on active **M**uscles
TA-M-NG	**N** and **G** electrodes on active **M**uscles
TA-M-PNG	**P**, **N** and **G** electrodes on active **M**uscles
TA-S-A	**S**hifting Electrodes: **A**rms up and down
TA-S-B	**S**hifting Electrodes: **B**end over
TA-S-C	**S**hifting Electrodes: **C**hest expansion
TA-S-J	**S**hifting Electrodes: **J**ump
TA-S-S	**S**hifting Electrodes: **S**quat
TA-S-W	**S**hifting Electrodes: **W**aist lateral twisting
BA-*: Biological Artifacts	
Abbr.	ECG Acquisition Artifact
BA-R-D	**R**espiration: **D**eep Breathing
BA-R-H	**R**espiration: Breath **H**olding
BA-R-R	**R**espiration: **R**apid Breathing

- Verify participant allergies to ensure compatibility with the electrode gel. Could you precisely position the electrodes at the clavicle and anterior superior iliac spine locations.
- Keep the data collection equipment (computer/laptop) at a maximum distance from the participant.
- Replace electrodes promptly to prevent overuse.
- Continuously monitor acquired signals to detect and address any unexpected electrode drops.

4.3 Dataset description

The h5-formatted dataset, comprising data from ten subjects, encompasses 199.82 minutes (3 hours, 19 minutes, and 49 seconds). Following the protocol, 163.50 minutes of the data are truncated and organized into 15, 30, or 60-second files, forming an accompanying set. Each file is labeled with the corresponding artifact abbreviation, serving as ground truth.

Efforts were made to maintain consistency in the quantity of data across various categories; however, specific categories with multiple samples were intentionally retained in the dataset to uphold transparency in the data collection process. While this decision aligns with the goal of openness and thorough representation, it resulted in a marginal imbalance in the distribution of the dataset. This approach was chosen to prioritize the comprehensive coverage of diverse scenarios within each category, contributing to the richness and versatility of the collected dataset.

The following summary of the dataset and participants uses the description in [LHS21] as a template.

4.3.1 Sensor placement

As in section 4.1 mentioned, the ECG signals were acquired using the Biosignalsplux ECG sensor [3], configured with low-noise ECG local differential triodes. The biosignalsplux ECG is primarily designed for single-lead ECG acquisitions in the Einthoven configurations. The Einthoven triangle allows you to acquire 3 different ECG leads which only depend on the positioning of your electrode cables[PLU20]:

- Lead I: measures from RA (right arm) to LA (left arm)
- Lead II: measures from RA to LF (left foot)
- Lead III: measures from LA to LF

Figure 4.1 shows the recommended electrode placement for the acquisition of ECG signals in Einthoven configurations depending on the different standard ECG sensors with difference electrode cable lengths [PLU20].

The position of three ECG electrodes are used in this study are:

- Positive electrode: Left clavicle, near left acromial end (LC).
- Negative electrode: Right clavicle, near right acromial end (RC).
- Grounding electrode: Left anterior superior iliac spine(LASIS).

ECG sensor electrodes positions and their measured muscles/body parts are listed as follows:

The left/right sternocleidomastoid muscles (when head turning to the opposite direction) and the quadriceps muscle (when seated) were used to simulate wrongly

[3] https://www.pluxbiosignals.com/products/electrocardiography-ecg-sensor-1 (accessed 26 Dec 2023)

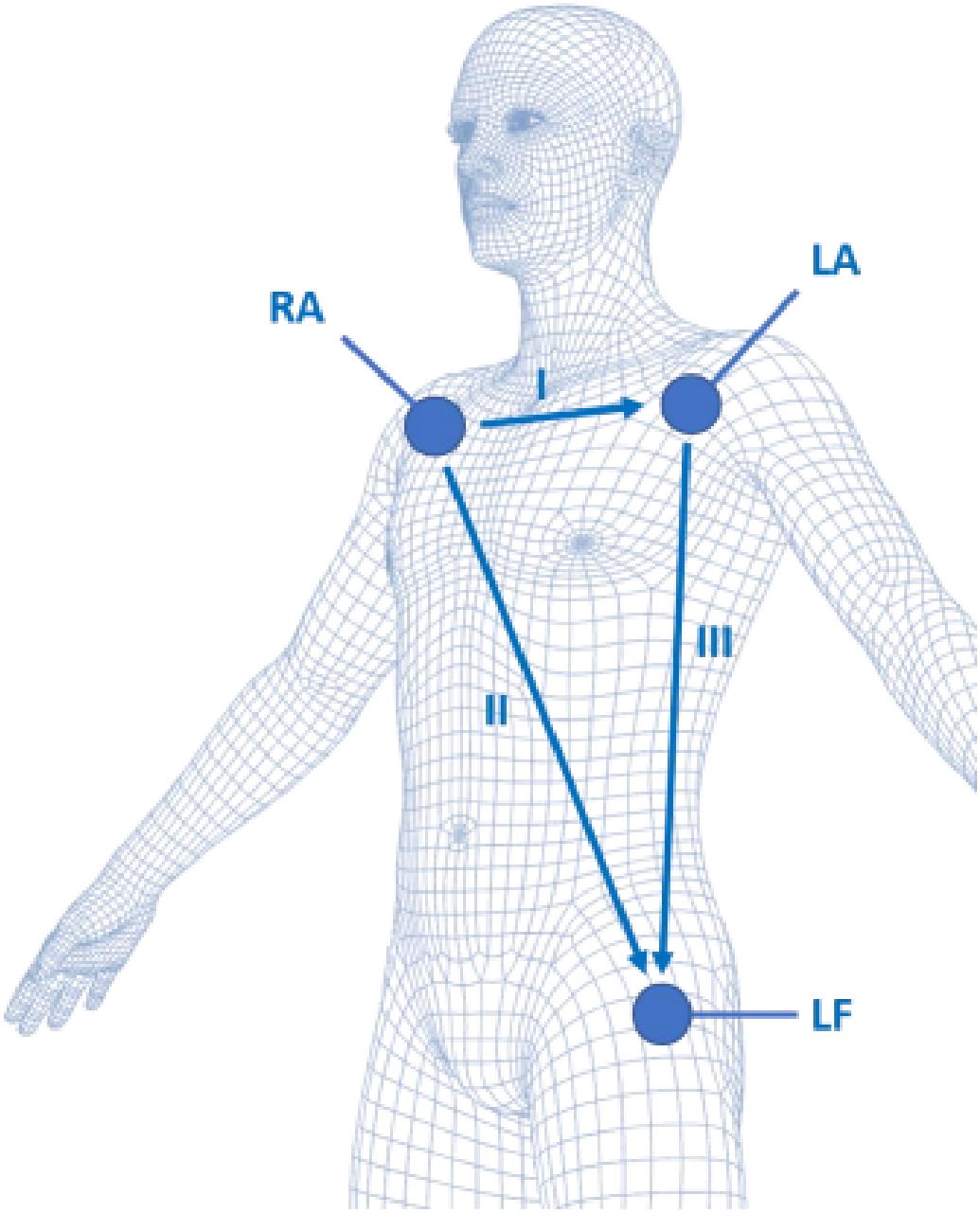

Figure 4.1: Electrode placements for ECG acquisitions in Einthoven configurations using the standard ECG sensor B with electrode cable lengths of 30cm (+, - & reference)[PLU20].

Table 4.2: ECG sensor electrodes position of various events. The following abbreviations are used in table. Positive electrods(+), negative electrodes(-), grounding electrodes(Reference). Left clavicle, near left acromial end(LC), right clavicle, near right acromial end(RC), left anterior superior iliac spine(LASIS).

Abbreviation	+	-	Reference
STD	LC	RC	LASIS
TA-*: Technical Artifacts			
TA-C-S	RC	LC	LASIS
TA-D-P	Detachment	RC	LASIS
TA-D-N	LC	Detachment	LASIS
TA-D-G	LC	RC	Detachment
TA-D-PN	Detachment	Detachment	LASIS
TA-D-PG	Detachment	RC	Detachment
TA-D-NG	LC	Detachment	Detachment
TA-E-IG	LC	RC	LASIS
TA-E-O	LC	RC	LASIS
TA-I	LC	RC	LASIS
TA-M-P	Left sternocleidomastoid	RC	LASIS
TA-M-N	LC	Right sternocleidomastoid	LASIS
TA-M-G	LC	RC	Right quadriceps
TA-M-PN	Left sternocleidomastoid	Right sternocleidomastoid	LASIS
TA-M-PG	Left sternocleidomastoid	RC	Right quadriceps
TA-M-NG	LC	Right sternocleidomastoid	Right quadriceps
TA-M-PNG	Left sternocleidomastoid	Right sternocleidomastoid	Right quadriceps
TA-S-A	LC	RC	LASIS
TA-S-B	LC	RC	LASIS
TA-S-C	LC	RC	LASIS
TA-S-J	LC	RC	LASIS
TA-S-S	LC	RC	LASIS
TA-S-W	LC	RC	LASIS
BA-*: Biological Artifacts			
BA-R-D	LC	RC	LASIS
BA-R-H	LC	RC	LASIS
BA-R-R	LC	RC	LASIS

placing the positive, negative, or grounding electrodes to active muscles (TA-M-*). When simulating the positive electrode sensing active muscles, the electrode was placed on the left sternocleidomastoid muscle, and the subject kept the head turned to the right, and vice versa. To simulate both the positive and the negative electrodes connecting to active muscles simultaneously, the subject kept the head tilted upwards to activate the sternocleidomastoid muscles on both sides[LZG+24].

4.3.2 Subjects

Ten participants, comprising four females and six males, ranging in age from 22 to 46 (mean age: 28.0 ± 6.9), were included in the data collection events. Each subject's involvement spanned approximately two hours, encompassing various activities such as announcements, Q&A sessions, equipment setup and adjustments, software preparation and test runs, comprehensive data collection following the protocol's categories, breaks, and the eventual release of the monitoring devices.

4.3.3 Ethic, privacy preservation and data security

Individual demographic information, encompassing age, gender, and general health status, has been meticulously documented for each participant. All participants provided written informed consent, and the study adhered to the ethical principles outlined in the Helsinki Declaration by the World Medical Association [Wor13]. In alignment with the consent form, the wearable sensor data was pseudonymized, ensuring the absence of any participant identification information in the dataset. The shared dataset is anonymized to uphold privacy and ethical standards. In addition, the consent form stipulates that the use of the data is limited to non-commercial research purposes, and the data users guarantee not to attempt to identify the participating persons. Furthermore, the data users guarantee to pass on the data (or data derived from it) only to third parties who are bound by the same rules of use (for non-commercial research purposes, no identification attempts, restricted disclosure). Data users who violate the usage regulation mentioned above will bear the legal consequences themselves, where the dataset publisher takes no responsibility[LHS21].

4.3.4 Data format

The dataset is manually organized into folders named by participants identifier, each dedicated to an individual participant and further categorized based on the type of activity. The root directory contains a total of 10 sub-directories named by "Subject_" and order name, representing the data of 10 subjects. Within each subdirectory, files follow a standardized naming convention, incorporating essential details such as participant identifier, signal type, acquisition date, artifact type, and session number. Simultaneously, the unaltered raw data, retaining the original file names, is systematically stored independently in a designated raw data folder, organized by participant number. This meticulous organization ensures easy accessibility and traceability of the dataset's components. The dataset is rewritten in HDF5 format to enhance user-friendliness, ensuring seamless compatibility with widely used data analysis tools. The modified HDF5 format has simple structure, which promotes effective utilization in various research and analytical contexts.

4.4 Data annotation

The dataset undergoes dual annotative methodologies tailored to different research scenarios. One annotative approach involves aligning the data with predefined events outlined in the protocol, primarily employed for identifying events that trigger artifacts. This annotation serves to furnish the user with targeted solutions for specific artifact-inducing events, empowering them to make necessary adjustments to the signal acquisition process based on the provided prompts to improve ECG signals quality.

Concurrently, an additional annotative method is implemented to pattern-based the ECG signal by scrutinizing the raw data in both frequency and time domains. This entails intercepting and annotating the signal based on distinct patterns observed during a session. For instance, an artifact induced by a jump may exhibit high-frequency noise patterns, EMG signal patterns, electromagnetic wave interference patterns, etc. This form of annotation serves to specifically identify artifact types. It is worth noting that while this annotation identifies the type of artifacts, it does not ascertain the root cause of the artifact. This second annotation contributes to reducing confusion arising from the simultaneous occurrence of multiple artifacts, as addressed by the first annotation.

4.4.1 Events-based

In the events-based annotation methodology, the dataset undergoes alignment with predefined events stipulated in the protocol, primarily serving the purpose of identifying events that trigger artifacts. Each event's start and end time correspond to the initiation and conclusion of a session. Given that session durations are predefined at 15s, 30s, and 60s, each annotated signal segment also aligns with these durations (15s, 30s, or 60s). The annotated labels adhere to abbreviations specified in the protocol, such as TA-E-O, TA-S-S, TA-M-G, etc.

Each annotated label encompasses crucial information:

- Type of Artifacts: Identifies the specific type of artifact induced.
- Location of Induced Artifacts: Specifies the location of induced artifacts, such as electrodes, muscles, etc.
- Cables of Induced Artifacts: Details the cables affected, including positive, negative, and grounding cables.

This annotation serves a practical purpose by offering tailored solutions for specific artifact-inducing events. For instance, if an electrode drops, the user is guided to reattach the electrode; similarly, if an electrode is incorrectly placed on a muscle, the user is instructed to adjust the electrode position to enhance the quality of the ECG signal. This user-friendly approach ensures that the calibration not only identifies artifacts but also provides actionable recommendations for mitigating and rectifying them during signal acquisition.

4.4.2 Pattern-based

In the pattern-based annotation methodology, initial identification involves annotating ECG signals that lack observability of QRST peaks in the time domain, classifying them as defected signals. Notably, defected signals are exempt from subsequent annotation refinement procedures. Non-defected signals undergo pattern interception and subsequent annotation to address diverse artifacts, including EMI artifacts, motion artifacts, high-frequency noise, EMG noises, absence of grounding, lead-off situations, powerline interference, inverted electrodes, overuse, insufficient gel, and baseline wander. Notably, powerline interference and high-frequency noise interference necessitate observation in the frequency domain for accurate assessment and classification. This approach ensures a systematic and comprehensive annotation process, distinguishing between defect and non-defect signals, and further refining non-defect signals by pattern-based annoation to address various artifacts.

The raw data from each session is meticulously examined using the Octave visualization tool with high resolution. Start and end time points of signal segments exhibiting the aforementioned pattern features are meticulously documented in a CSV file, each labeled with the corresponding pattern label. In the subsequent signal processing stage, signal segments are selectively intercepted based on these recorded start and end time points. Segments within the same session devoid of artifacts and noise are appropriately labeled as standard ECG signals. This systematic approach ensures precise identification and isolation of signal components for further analysis.

Due to constraints in professional human resources, the pattern-based annotation was undertaken by a singular individual and not cross-validated by multiple experts. To enhance the robustness of the annotation and ensure accuracy and reliability, future work may involve the involvement of multiple individuals in the annotation process. This iterative approach is crucial for minimizing potential biases and enhancing the overall validity of the calibration process, thereby mitigating any adverse effects on subsequent identification processes.

5. Data Preparation

Data preparation serves as a foundational and critical step in the machine learning pipeline, significantly influencing the subsequent modeling and analysis processes. In this study, the objective is to achieve artifacts classification through machine learning models, necessitating the preprocessing of raw data. The utilization of the entire signal for training models is computationally expensive and inefficient. In the study, the ECG signal has a sample rate of 1000Hz, resulting in a substantial number of samples (60000) for each events-based STD ECG. However, not every sample is equally meaningful for signal representation. The inclusion of redundant information not only exponentially increases computational demands but also introduces interference into the machine learning process, potentially leading to classification confusion.

To provide more representative data to the models, feature selection assumes a pivotal role in artifacts classification. In the current research stage, which exclusively involves non-deep learning methods, features are manually extracted using the Time-Series Feature Extraction Library (TSFEL) [BFF+20]. All experiments have been rigorously conducted using leave-one-out cross-validation (LOOCV) for person-independent training. This comprehensive validation approach ensures the robustness of offline person-independent models, with the overall recognition rate calculated as a macro average by aggregating results from ten-fold experiments. Further details about person-independent models will be expounded upon in Section 6.

Three non-linear models are trained with the same dataset and data representation. Non-linear models are preferred due to their flexibility and superior fitting capabilities on complex datasets compared to linear models. The dataset exhibits varying pattern characteristics among different artifacts, suggesting that classification within distinct artifact types is intricate. Consequently, assuming a linear relationship between features extracted by TSFEL becomes challenging, underscoring the rationale for selecting non-linear models as the primary approach in this study.

5.1 Data segmentation with sliding windows

Sliding windows are a fundamental technique in data segmentation, serving as a dynamic tool to partition continuous biosignals data into computationally friendly

segments. This approach involves systematically moving a length fixed window across the dataset, creating overlapping or non-overlapping segments based on customizable parameters such as window length and overlap ratio. Sliding windows play a crucial role in temporal localization, facilitating the identification of specific events or patterns within biosignals. The overlap between adjacent segments ensures smooth transitions and captures gradual changes in the signal. Often used in conjunction with feature extraction, which is introduced in subsection 5.2, sliding windows enable the characterization of temporal patterns, enhancing the adaptability and granularity of data segmentation for tasks such as time-series analysis, event detection, anomaly identification.

A standard ECG signal has a recognizable PQRST waveform within a reasonable duration. However, diverse artifacts may deviate from the correct ECG waveform and expected typical duration. In the context of strictly periodic ground signals, repetitive signal features align with events, while the appearance of non-periodic signal features holds significance for identification. Unlike a standard ECG signal that exhibits clear and perfect repetition of PQRST peaks, an ECG signal with artifacts may obscure or deviate from the standard waveform, with artifacts potentially extending into other periodic waveforms like EMG artifacts. Segmenting periodic signals proves valuable as it preserves and emphasizes non-periodic signal features, easily discernible by machines when the similarity to other segments, especially standard ECG segments, is low. Modulating the window length enables targeted recognition of periodic features associated with varying cycle lengths. For instance, a longer window enhances the recognition of periodic EMG artifacts, characterized by cycle lengths relative to high-frequency artifacts. Segmenting artifacts with relatively stable performance increases their proportion in the data recognition pool, thereby enhancing the machine's ability to learn the characteristics of this specific feature subset.

Given the diverse adaptability of different artifacts to various window lengths, it remains challenging to definitively assert the superiority of a specific window length for universal artifact recognition. Acknowledging that window length and overlap jointly influence the features encapsulated in a signal fragment, we undertook a comprehensive exploration involving both parameters. The investigation spans window lengths ranging from 1s to 9.5s with a 0.5s step size, and overlaps ranging from 0% to 90% with a 10% step size. A rigorous training regimen encompasses all possible combinations of window lengths and overlaps. This exhaustive approach aims to identify a combination with optimal accuracy, which could subsequently be applied in an online demonstration setting.

5.2 Significance of features extraction for machine learning

Features extraction is a process of reconstruction, where raw data is converted to a feature-described representation. Raw data normally has a large volume of information but also has irrelevant, misleading, and redundant data. Processing raw data costs a lot of computation resources which requires strong computational power or relatively long computation times to achieve. The sampling rate of the database utilized in this study is 1000 Hz. For instance, within 1 second, all 1000 samples

Temporal	Statistical	Frequency
Absolute Energy	Maximum	Entropy
Area under the curve	Mean	Fundamental Frequency
Autocorrelation	Minimum	Max frequency
Centroid	Standard deviation	Roll off
Distance	Interquartile range	Roll on
Mean absolute difference	Kurtosis	Spectral distance
Mean difference	Mean absolute deviation	Spectral kurtosis
Sum of absolute difference	Root mean square	Spectral skewness
Total energy	Skewness	Spectral spread
	Variance	
	Median	

Table 5.1: All 30 used features in each domain from TSFEL library[RLF+22]

cannot be utilized as features for artifact recognition. Moreover, many individual samples, without processing, exhibit similar performance across different artifacts. Therefore, raw data necessitates processing to extract decisive data feature points essential for artifact classification. This approach aims to achieve accurate artifact classification.

Since ECG signals are time-series signals, the TSFEL Python software package was used in the feature extraction process. TSFEL automatically extracts more than 60 different features in the statistical, temporal, and spectral domains. For more details on specific features and source code, please refer to [BFF+20] [1]. [Liu21][HLS21][NS20][GVR+22][GCS+22] have proven that TSFEL performed effective and efficient on processing various biosignals, such as EMG, ACC, EEG and audio signals etc. 50% of all TSFEL features in the statistical, temporal, and frequency domains with relatively lower computational costs are selected over, as listed in Table 5.1, regarding the real-time computing.

During the initial training phase, all 30 features were extracted and utilized for training, as the distinguishable features between different artifacts were yet to be apparent. Manual selection of features through experience or observation needs to have conviction. Features for distinguishing between different types of artifacts should exhibit machine-learning-ready variability when various artifacts are encountered. Following the machine learning model's determination, feature selection is undertaken using time-domain, frequency-domain, statistical-domain, and greedy algorithms to enhance classification accuracy and efficiency and minimize computational delay during real-time classification. This process aims to retain the most distinguishable features among artifacts from the pool of 30 features.

[1] https://tsfel.readthedocs.io/en/latest/ access 10.Feb.2024

6. Experiments, results, and discussions

This section is divided into four parts: person-dependent (PD) experiments, semi-person-independent (semi-PI) experiments, and person-independent (PI) experiments. Each experiment evaluates the performance of three classifiers (DT, SVM, RF) during training. The final section presents a pattern-based PI experiment to investigate whether the model's performance is impacted when the event-based class includes patterns of multiple types of artifacts.

Results from the PD experiment validate the experimental design, especially the protocol formulation (refer to Section 6.1 for detailed results). Transitioning from PD to PI experiments, a semi-PI experiment was introduced in section 6.2. The effect of individual differences on artifact recognition can be discerned by comparing its outcomes with those of the PI experiment. Implementing semi-PI experiments can enhance the recognition accuracy of standard ECG signals, making them more applicable in practical scenarios.

In the PI experiments, a vertical layered architecture in Figure 6.1 is proposed to enhance the model's performance. This involves prioritizing different classes during categorization. For example, in a one-tier architecture, all classes are processed simultaneously. In the 2-M architecture, binary classification of standard ECG (STD) and ECGs with artifacts is initially conducted, followed by artifact classification for ECGs with various artifacts. The hybrid architecture(H-M) combines 1-M and 2-M. In contrast to 2-M, H-M enables that even segments judged as artifacts in the first layer could still be classified as STD in the second layer, with a view to further reducing the number of STD's false negative instances [LZG+24]. Another rationale for suggesting a hierarchical structure is the significant disparity in the database's composition, with the proportion of standard ECG data recording from sitting down and standing up being six times higher than other individual classifications. Such an imbalance could lead to a bias towards standard ECG data in the classification results, a phenomenon particularly pronounced in a one-tier structure.

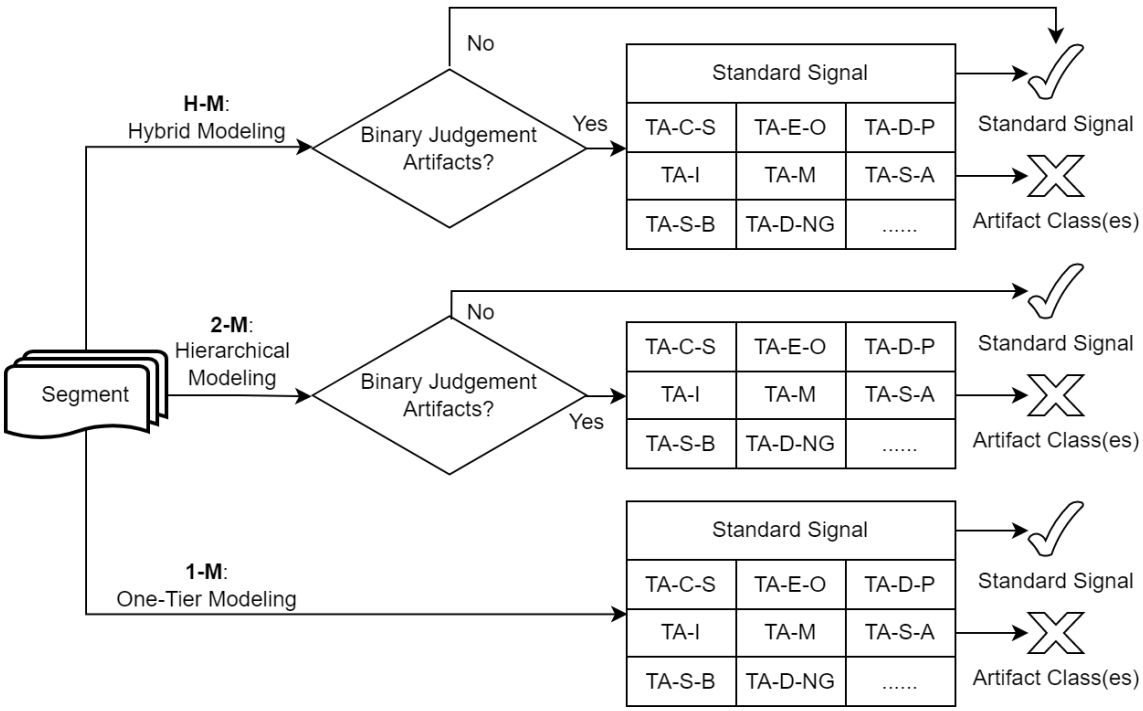

Figure 6.1: The flow charts of the one-tier modeling (bottom), the hierarchical modeling (middle), and the hybrid modeling (top)[LZG+24].

The study focuses primarily on one-tier architecture and the preferential binary classification experiment proposed by 2-M. The enormous potential utility of 2-M and HM in real-time models cannot be overlooked, especially when introducing sequential modeling or deep learning in future experiments, or when combining pattern-based and event-based recognition.

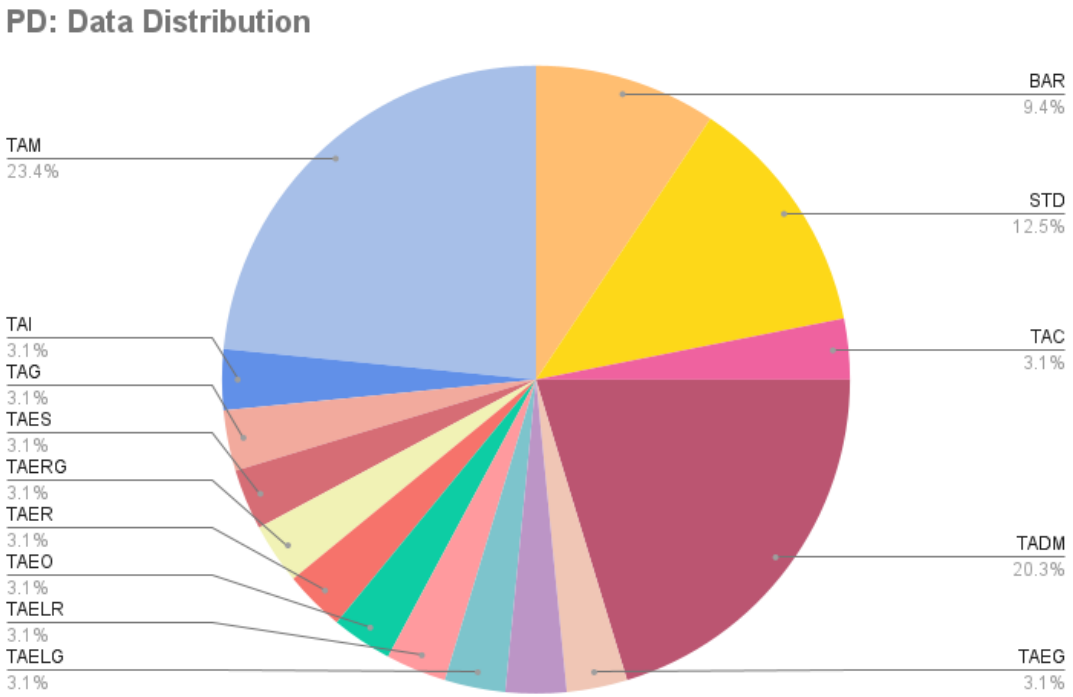

Figure 6.2: Small-scale dataset of person-dependent (PD) and semi-person-independent experiments (semi-PI).

In PD and semi-PI experiments, abbreviations remain the old abbreviations. A list of old abbreviations follows 6.1. In addition, these two experiments were applied to a small-scale database consisting of six subjects, whose data distribution is shown in Figure 6.2.

In this study, the majority of default settings for three models, decision tree (DT), support vector machine (SVM), and random forest (RF), were employed from the *Python* package *scikit-learn*. The following listed specific parameters, determined through iterative experimental parameter tuning validation, remained consistent across PD, semi-PI, and PI training.:

- DT: random state = 0.
- SVM: radial basis function (RBF) kernel with c = 80 and gamma = 250.
- RF: min samples split = 2 and n estimators = 25.

6.1 Person-dependent experiments

The PD experiments utilized a database of 6 individuals. A 4-fold cross-validation of the training results was employed to ensure the robustness and generalization ability of the model. Focused on a one-tier model architecture, the experiments aimed to consolidate confusable classes to enhance the model's classification performance. Initially, the optimal combination of window length and overlap ratio was identified. Subsequently, feature selection was conducted for each of the three feature domains—statistical, temporal, and frequency—based on this optimal combination.

Table 6.1: Taxonomy of real-time artifacts during ECG acquisition: old abbreviation

TA-*: Technical Artifacts	
Abbreviation	ECG Acquisition Artifact
TA-E-S	**E**lectrodes **S**witched: positive and negative
TA-E-L	**E**lectrodes: **P**ositive electrode
TA-E-R	**E**lectrodes: **N**egative electrode
TA-E-G	**E**lectrodes: **G**rounding electrode
TA-E-LR	**E**lectrodes: **P** and **N** electrodes
TA-E-LG	**E**lectrodes: **P** and **G** electrodes
TA-E-RG	**E**lectrodes: **N** and **G** electrodes
TA-G	Insufficient **G**el
TA-E-O	**E**lectrodes **O**verused: **P**, **N**, and **G**
TA-I	Magnetic field **I**nterference
TA-D-M	**D**etachment on **M**uscle
TA-M	Shifting electrodes: Body **M**ovement
TA-C	**C**able: short cable
BA-*: Biological Artifacts	
Abbr.	ECG Acquisition Artifact
BA-R	**R**espiration: Deep Breathing, Breath Holding, Rapid Breathing

The results obtained from the PD model assessed the discriminability among artifacts, thereby validating the study's artifacts simulation protocol and experimental design.

	window length	overlap ratio	accuracy	f1-score
DT	6.0s	0%	56.45%	0.55
SVM	9.0s	60%	59.34%	0.56
RF	9.5s	20%	59.48%	0.56

Table 6.2: The most optimal combination of window length and overlap ratio for decision tree (DT), support vector machine (SVM) and random forest(RF)

6.1.1 Window length and overlap ratio selection

In the PD experiments, window length and overlap ratio were integrated into a comprehensive evaluation framework. Window lengths ranged from 1s to 9.5s in increments of 0.5s, resulting in 19 distinct lengths. Similarly, overlap ratios varied from 0% to 90% in increments of 10%, totaling 10 overlap ratios. This combination resulted in a total of 190 unique combinations of window lengths and overlap ratios. To identify the optimal combination, the classification accuracy and f1-score across these combinations was compared during the evaluation process. The higher the f1-score, the more robust the model performance.

Following separate utilization of features within the time, statistical, and frequency domains, the decision was made to prioritize features from the comprehensive statistical domain, as elaborated in subsection 6.1.2.

The optimal combinations of window length and overlap ratio for the three classifiers are displayed in the table 6.2 below. It is evident that all three classifiers tend to perform better with longer window lengths, particularly 6s, 6.5s, and 9s. RF, in particular, achieves the highest accuracy with a window length of 9.5s and overlap of 20%. While longer window lengths show minimal sensitivity to overlap ratio, overlap ratio tends to improve classification accuracy with shorter window lengths. Overall, window length emerges as the predominant factor affecting accuracy.

	Statistical Domain	Temporal Domain	Frequency Domain
DT	39%	18%	15%
SVM	45%	18%	11%

Table 6.3: Classification accuracy of decision tree (DT) and support vector machine (SVM) across statistical, temporal and frequency feature domains.

6.1.2 Feature selection

As depicted in the table 5.1, tsfel offers 30 features extracted from the time, statistical, and frequency domains. These features were grouped into statistical, frequency and temporal domains during the experiments. The table 6.3 showcases the highest accuracy achieved by the decision tree and SVM classifiers using various features, window lengths, and overlap ratios.

With features from the statistical domain, DT achieves an accuracy of 39%, while the SVM classifier achieves an even higher accuracy of 45%. This accuracy is twice as high as when using features from the time domain alone and three times as high as when using features from the statistical domain alone. These results indicate that different types of ECG artifact types are more distinguishable in the statistical domain. Both temporal and spectral domain features have been shown in various biosignal research areas, but most research tasks are based on (as far as possible) correctly acquired signals containing advantageous time (e.g., zero-crossings and slop) and frequency (e.g., periodicity) characteristics. On the contrary, most signals studied in this article are exceptional because they are irrational and do not conform to everyday patterns, which may explain the powerlessness of features in temporal and frequency domains in the pilot study [LZG+24]. Consequently, the focus of subsequent experiments shifts to 11 statistical domain features.

Both forward and backward greedy algorithms were employed for feature selection within the 11 statistical domains. Following the determination of the optimal combination of window length and overlap ratio for the three classifiers, the forward greedy algorithm initiates with a single feature. It progressively adds features from the statistical domain, selecting those resulting in the highest classifier accuracy. This iterative process continues, incrementally adding one feature at a time, until all possible combinations are evaluated to identify the optimal feature set for each classifier.

Conversely, the backward greedy algorithm begins with all 11 features. It calculates classifier accuracies and systematically removes features, retaining only those contributing to the highest accuracy. Similarly to the forward approach, features are removed one at a time, and accuracies are recalculated for each combination until the most optimal feature set is determined.

By combining the results of both algorithms, the most desirable combinations of statistical domain features are obtained. Subsequent experiments utilized 11 statistical features to train the classifier.

6.1.3 Classification results

Figure 6.3 and 6.4 present the classification results of SVM for the PD experiments. The table 6.4 reveals that SVM achieves an accuracy of 52.82% using full statistical domain features, with a window length of 9 seconds and an overlap ratio of 60%. Similarly, RF achieves an accuracy of 52.85%, comparable to SVM but with faster computation. Additionally, the confusion matrix depicted in the figure below illustrates SVM's classification results for each class, with "BA-R" representing the Breathing class, which is often confused with TA-G (Detailed discussion in subsection 6.1.4).

More notably, results in figure 6.3 showed, that STD is misclassified as BA-R and TA-G, indicating a significant failure in one of the most crucial aspects of recognition. Such misclassification can lead to users consistently receiving prompt messages during real-time recognition, thereby interfering with their normal acquisition of ECG signals. To enhance the recognition rate of STD, an attempt was made to merge similar classes. However, upon observing BA-R, it was discovered that the artifact caused by breathing was minimal, making it challenging to eliminate the artifacts introduced during signal collection. Therefore, combining BA-R and STD could be considered without disrupting the regular use of signal analysis.

Another approach involves treating the BA-R class separately. While validating the feasibility of real-time artifact recognition, BA-R is initially excluded from the training and test sets. In future iterations, once the real-time performance of the model recognition stabilizes, BA-R will be incorporated for classification. The table 6.5 presents the original STD recognition accuracy alongside the STD recognition accuracy corresponding to the two proposed methods aimed at enhancing the STD recognition effect.

The recognition accuracy of STD is notably improved among all classifiers after removing BA-R. RF and DT achieve a 100% recognition accuracy. The recognition accuracy of STD with SVM increases significantly when BA-R and STD are merged. The performance of the other two classifiers remained the same. This may also be attributed to the need for fine-tuning of the classifiers. Temporarily removing BA-R from the database can significantly enhance STD recognition rates.

The artifactual features of TA-D-M are notably intricate, involving various simulated movements such as chest expansion, arm lifting, squatting, forward body flexion, jumping, and crotch twisting during ECG collection. This complexity is evident in the classification results, where TA-D-M may be mistaken for multiple classes including STD, TA-E-S, TA-G, and TA-I. The artifacts produced by these overused electrodes can be mistaken for standard ECG signals. In the figure 6.4 illustrating window lengths and overlap ratios, no consistent trend was discerned. Instead, certain specific window lengths and overlap ratios notably enhanced the performance of the classifiers.

Classifier	Window length	Overlap ratio	Feature	Accuracy	f1-score
DT	6s	0%	all statistic	45.61%	0.45
SVM	9s	0.6%	all statistic	52.82%	0.48
RF	9s	20%	all statistic	51.49%	0.49

Table 6.4: Best accuracy of decison tree (DT), support vector machine (SVM), random forest (RF) with parameters list in person-dependent(PD) experiments (window length, Overlap ratio, Features, Accuracy, f1-score)

Classifier	All label	Merge BA-R and STD	Remove BA-R
DT	0%	0%	100%
SVM	0%	100%	62%
RF	0%	0%	100%

Table 6.5: Accuracy of STD recognition with dataset in three handling of BA-R class (no handling, merged BA-R and STD, removed BA-R)

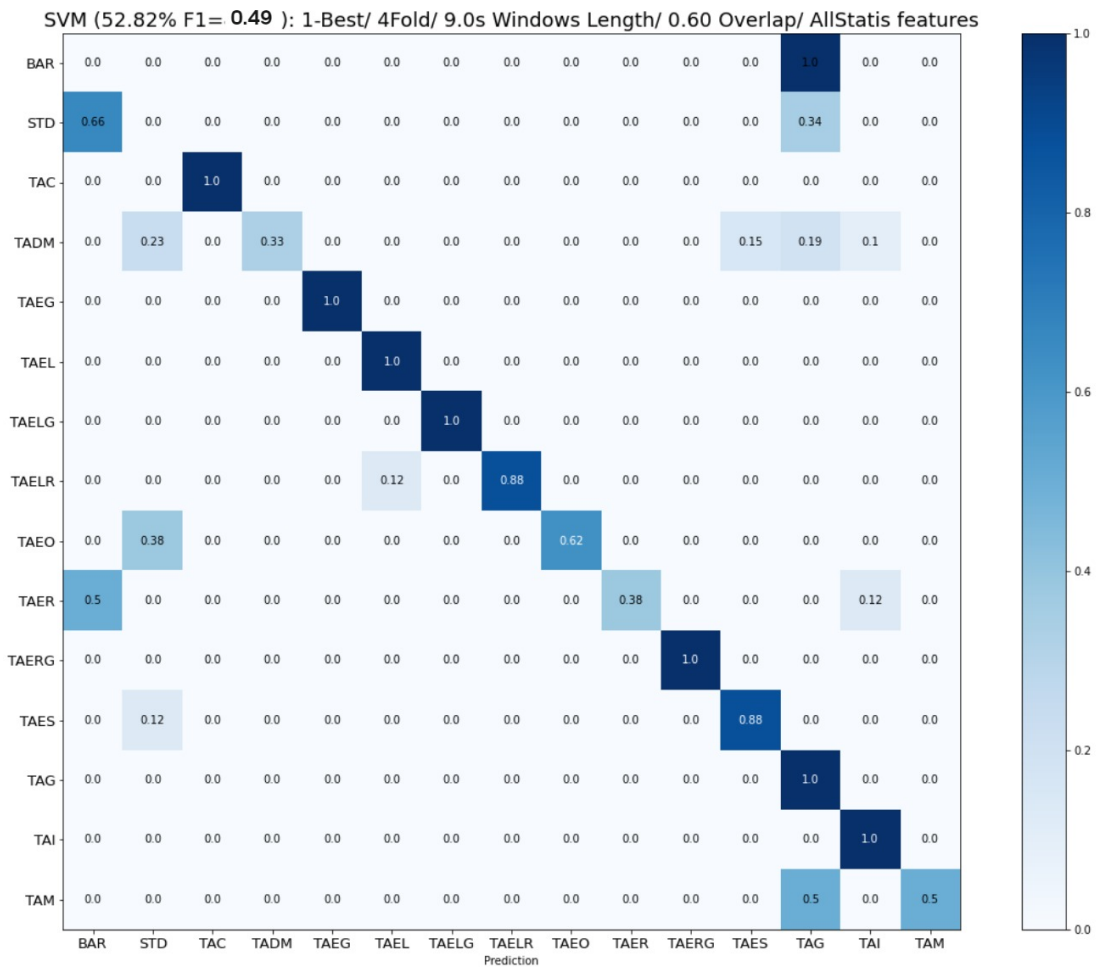

Figure 6.3: Confusion matrix of the one-tier support vector machine (SVM) modeling with 11 statistical domain features, 9-second window length, 60% overlap ratio, person-dependent 4-fold cross-validation and 1-best output.

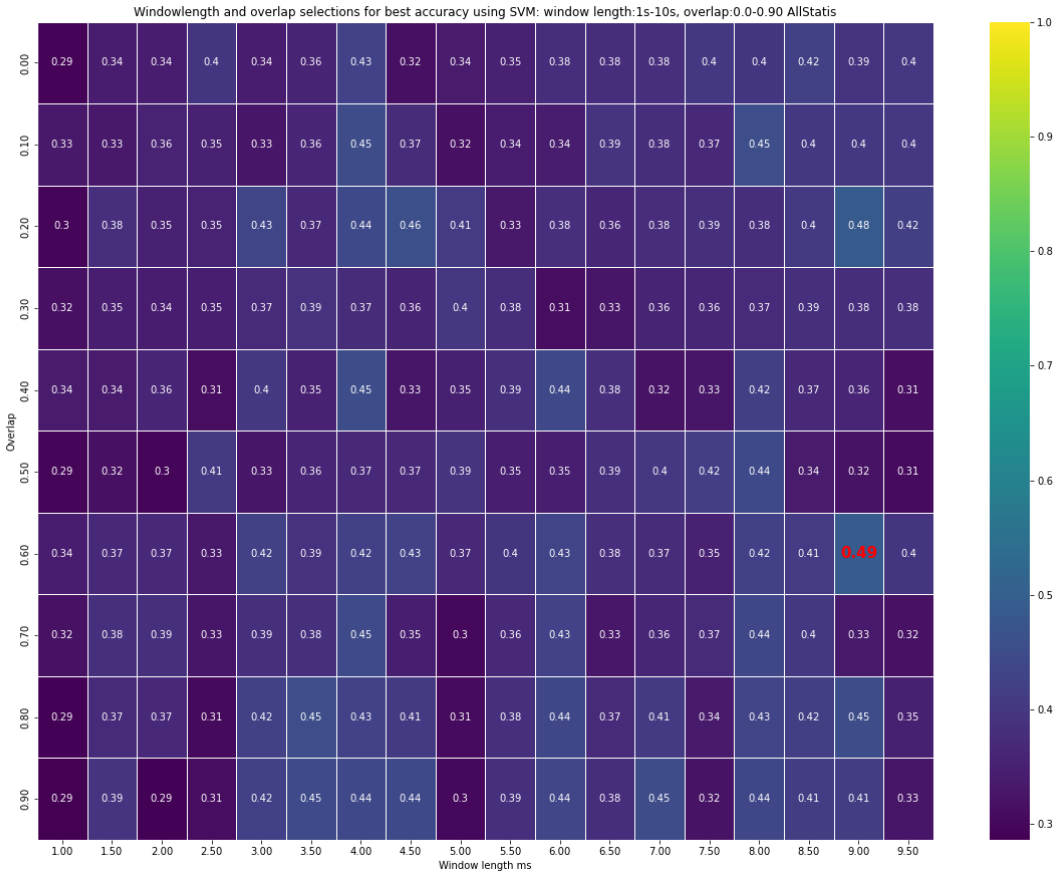

Figure 6.4: Heatmap of all combinations of window length and overlap of the one-tier support vector machine (SVM) modeling with 11 statistical domain features, person-dependent 4-fold cross-validation and 1-best output.

6.1.4 Discussion

Figure 6.5 illustrates the signals of the TA-G in the case of insufficient analog gel. The isolation paper of the electrodes was divided equally into eight parts to simulate 12.5%, 25%, 37.5%, 50%, 62.5%, 75%, and 87.5% of the gel being masked. Overall, the masked signals closely resembled the original signals, with many appearing even smoother, as if a low-pass filter had been applied. This effect is particularly noticeable when the masked portion does not exceed 50% (see Figure 6.5.

Up to 87.5% (7/8) of the electrodes were masked because further masking would have made it difficult to attach the electrodes to the skin. During data recordings from different subjects, electrodes were masked by 25%, 50%, 75%, or 87.5% to simulate various gel deficiency scenarios (masking rates are recorded in the file name). Tests were conducted by shielding each pair or all positive, negative, and grounded electrodes. The results showed no significant differences in signaling patterns, leading to the capture of signals only from shielded grounded electrodes [LZG+24].

This smoothing effect results in BA-R, TA-G, and STD being extremely similar from an observation. The confusion matrix reveals that BA-R and STD are often mistaken for TA-G.

The artifactual features of TA-D-M are notably intricate, involving various simulated movements such as chest expansion, arm lifting, squatting, forward body flexion, jumping, and crotch moving during ECG collection. This complexity is evident in the classification results, where TA-D-M may be mistaken for multiple classes including STD, TA-E-S, TA-G, and TA-I. Attempts were made to categorize artifacts based on different movements, yet the classification outcomes remained unsatisfactory. Moreover, differences in artifact amplitude caused by individual variations may lead to discrepancies within signals of the same class during signal acquisition. Presently, there is no ideal method to ensure consistency in motion amplitude during the signal collection process. Furthermore, controlling the user's motion amplitude in real-time applications poses significant challenges and will be addressed in future work.

Equally poorly controlled are overused electrode pads. The artifacts produced by these overused electrodes can be mistaken for standard ECG signals, suggesting a close resemblance between the simulated overused electrode signal and the characteristics of a standard ECG signal, likely due to mild overuse. Research into standardizing the artifacts characteristics resulting from overuse, or enhancing the generalization of such artifact and extracting distinguishable features from it, could be pursued in future studies.

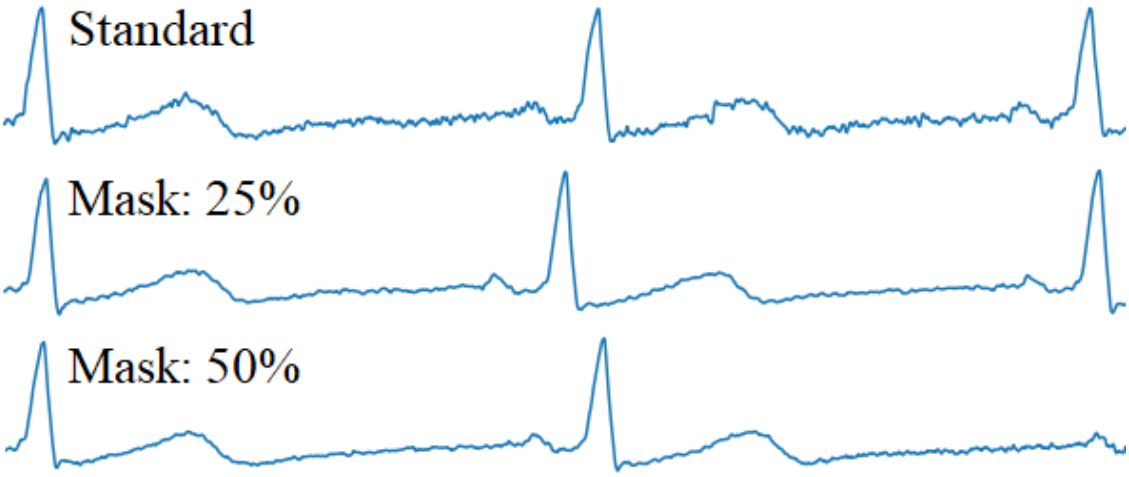

Figure 6.5: Example of masking electrodes to simulate the gel deficiency situation. Top: standard electrode; middle: 25% of the gel is covered by a right-angle sector; bottom: 50% of the gel is covered by two right-angle sectors[LZG+24].

6.2 Semi-Person-Independent experiments

During the transition to PI experiments, an attempt was made to further enhance the recognition accuracy of STDs. This involved testing the utilization of 50% of the STD from the currently recognized subject for training, while reserving the remaining 50% for evaluation. Additionally, STD from other individuals were incorporated into the training process, while ensuring that all artifact categories remained completely person-independent. This approach, known as semi-PI training, requires experimenters to pre-record user signals for tens of seconds in a standardized manner.

6.2.1 Classification results

The table 6.7 presents a comparison of the overall recognition accuracy and STD recognition accuracy for the three classifiers in both PD and semi-PI experiments. The recognition accuracy represents the highest achieved among various combinations of window length and overlap ratio utilizing 11 statistical features.

In PD experiments, the classifier's performance consistently outperforms that of the PI experiments, a trend that continues in the proposed semi-PI experiments. It's evident that all three classifiers exhibit lower recognition accuracy for all classes (except BA-R) in the semi-PI experiments compared to the results of PD experiments. However, incorporating 50% of the subject's own STDs in the training set maintains high accuracy in STD classifications despite a decrease in recognition accuracy compared to the recognition accuracy of PD experiments for most other classes.

Section 6.3.2 will further compare the recognition accuracy with the PI model.

6.2.2 Discussion

In the 1-M semi-PI experiment, the performance notably lags behind in the PD experiments, particularly affecting SVM's accuracy. The table's first two columns

Table 6.6: Recognition rates of semi-person-independent(semi-PI) experiments with the random forest (RF) and the support vector machine (SVM) modeling for comparing the performance of one-tier pure person-dependent (PD) with 4 folds and semi-person-independent (semi-PI) with 7 folds training with 11 statistical domain features. semi-PI values inferior to PD are indicated in gray.

Classifier	**STD+T: PD**	semi-PI	**STD: PD**	semi-PI
DT	58.04%	48.19%	100%	74%
SVM	66.16%	48.10%	62%	78%
RF	63.31%	49.62%	100%	77%

show the average recognition accuracy of PD and semi-PI across both STD and all T classes, with the last two columns indicating their performance specifically in the STD class. Gray shading highlights instances where semi-PI's accuracy falls short of PD. Moreover, semi-PI's experiment employed a 7-fold cross-validation, while PD used a 4-fold one, reflecting the database's expansion with more subjects.

Results indicate that SVM achieves higher accuracy in STD classification in the semi-PI experiment than PD. However, PD consistently outperforms semi-PI across other experiments due to inherent individual variations in artifacts. For instance, differences in movement and amplitude during artifact collection lead to variations like TA-M. Currently, there's no practical method to control these individual differences.

The undesirable result is, while SVM's STD classification accuracy improves in the semi-PI experiment with the addition of 50% STD data from test subjects, reducing confusion between STD and TA-E-O and TA-M. TA-D-M also showed similar classification results. However, the recognition accuracy of other classes in the PD experiment is higher than that in the semi-PI experiment, so the above situation is accidental and not representative.

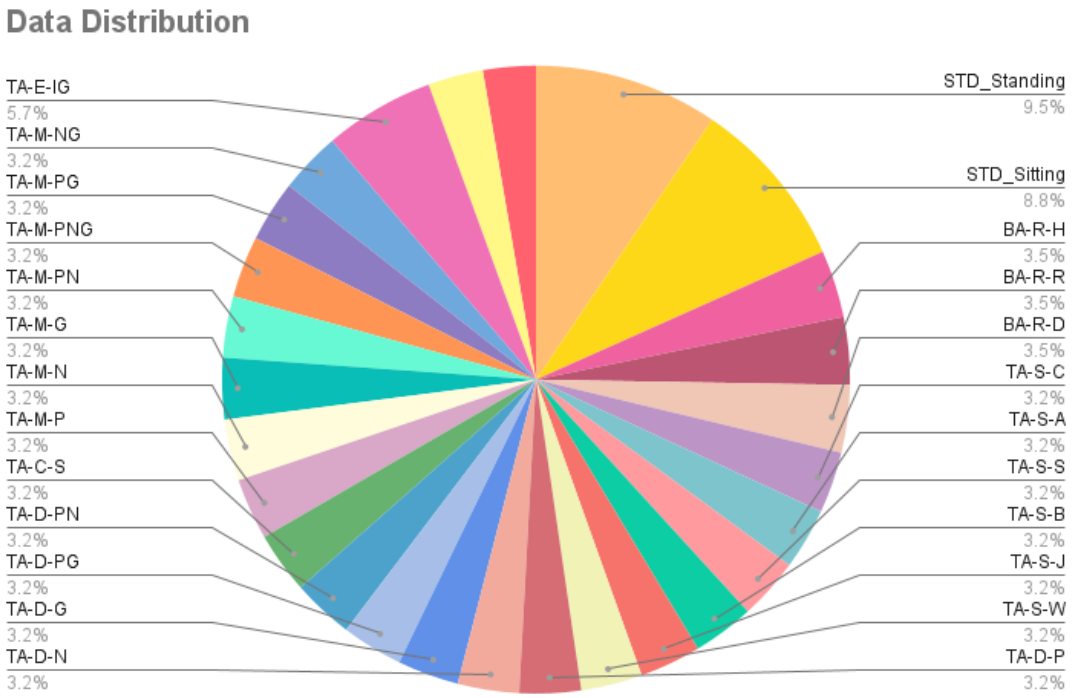

Figure 6.6: Data distribution of 10-subject dataset.

6.3 Person-Independent experiments

This subsection delves into the results of the PI experiments, comparing them with those of semi-PI and PI experiments. Additionally, it analyzes the outcomes of experiments distinguishing between the one-tier model and prioritizing STDs in the model with 2-M . All PI experiments were conducted on an 11-person database, with the data distribution detailed in figure 6.6. The abbreviations used are the new version of the acronyms, which are described in detail at table 4.1. The data distribution of the other classes is relatively balanced, except for STD, where the data size of STD is six times larger than the rest of the sample. The classifier remained focused on utilizing 11 statistical features, with parameter choices for window length and overlap ratio.

6.3.1 Classification results and discussion

One-tier and hierarchical modeling

A one-tier architecture treats STD and other artifact types equally for recognition purposes. Conversely, a layered architecture transforms STD recognition into binary recognition, distinguishing between STD and ECG with artifacts. As previously mentioned, a lower STD recognition accuracy can result in frequent cue messages sent to the user, disrupting proper signal collection. Semi-PI experiments, which involve collecting STD signals from subjects in advance to serve as a reference for recognition, are feasible. However, a fully PI model with a high recognition accuracy can streamline the use of real-time models and enhance the user experience. Therefore, the concept of 2-M/H-M architecture is proposed here to prioritize the recognition accuracy of STDs by simplifying the classification approach and emphasizing the distinguishing features of STDs.

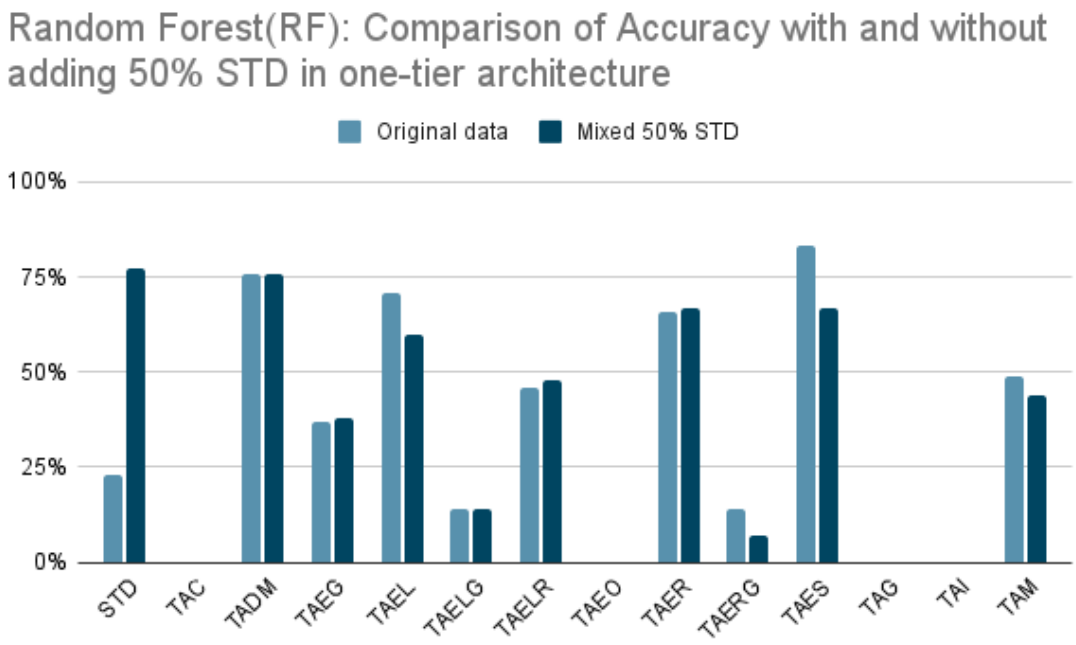

Figure 6.7: Comparison results: recognition accuracy with and without 50% STD, with random forest classifier, 1-Best, 7-Folds, 5s window length, 0% Overlap ratio, 11 statistical domain features, person-independent leave-one-out cross-validation (LOOCV), and one-tier architecture

Figure 6.7 shows the recognition accuracy of STDS with the addition of 50% of testers and all classes without the addition in a single-layer structure model. The recognition accuracy of STD was increased from 23% to 77%, indicating that this method effectively improved the model's performance in recognizing STD. Comparing the performance of the other classes, it can be seen that adding the STD of the tester did not significantly help the recognition of the different classes and even had a negative effect on the recognition of TA-E-L and TA-E-S. This may have something to do with the signal quality of the added STD, and it also reflects the confusing relationship between STD, TA-E-L, and TA-E-S. The 11 statistical domain features used in the experiment didn't represent the differences between these three categories well.

Illustrating the recognition accuracy of STDs in RF, figure 6.7 showcases a notable 37% enhancement in recognition results with the 2-M architecture compared to the one-tier architecture. Remarkably, when coupled with 50% of the test set subjects' own STD samples, the 2-M architecture demonstrates the highest performance, achieving a recognition accuracy of up to 82%. This comparison underscores the effectiveness of the 2-M architecture in significantly improving the recognition accuracy of STDs.

Figure 6.8 compares the recognition accuracy of different types in different structural models. As a priority in the experiment, STD was first binary classified; that is, all ECG with artifacts were grouped into one class, STD, and 50% of the test subjects' STDs were added into the same class for binary identification. All ECGs with artifacts were then identified by artifact type. This simulates a 2-M model. It can be seen from the results that the recognition rate of STD in the 2-M structure is 4%

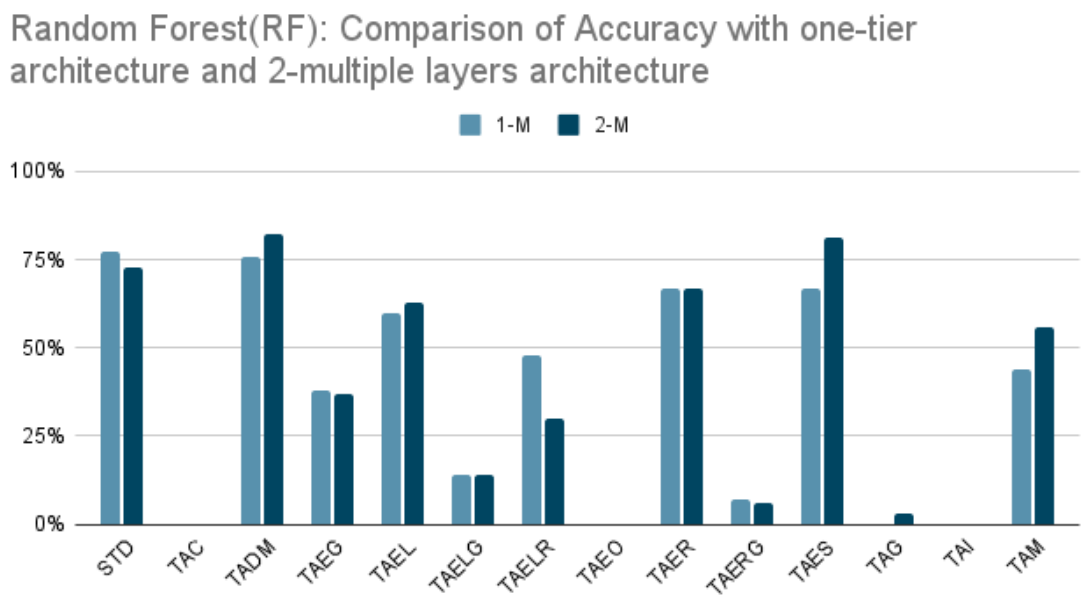

Figure 6.8: Comparison results: recognition accuracy with one-tier (1-M) architecture and with two multiple layers (2-M) architecture. 1-M modeling with random forest classifier, 1-Best, 7-Folds, 5s window length, 0% Overlap ratio, 11 statistical domain features, person-independent leave-one-out cross-validation (LOOCV), and one-tier architecture. 2-M modeling: STD vs. T layer with random forest classifier, 1-Best, 7-Folds, 3.5s window length, 0% Overlap ratio, 11 statistical domain features, person-independent leave-one-out cross-validation (LOOCV). T layer with random forest classifier, 1-Best, 7-Folds, 8.5s window length, 60% Overlap ratio, 11 statistical domain features, person-independent leave-one-out cross-validation (LOOCV)

lower than that in the single-layer structure, which may be directly related to the data distribution. In binary classification, the sum of data with artifacts is about nine times that of STD data, so the recognition result must be skewed like the data with artifacts. For artifact recognition results in the second layer, the two-layer structure improves the performance of TA-D-M, TA-E-L, TA-E-S, and TA-M. This shows that 2-M/H-M model results might help improve the recognition accuracy and reduce the confusion brought by STD on other artifact types' recognition.

The current 82% STD recognition accuracy indicates the need for improvement, as achieving a high accuracy for STD recognition is crucial for establishing a dependable foundation for the real-time model. Consequently, efforts to address the recognition of other artifacts are prioritized only once there is no disruption to signal collection caused by misclassified STDs. However, the current state of imbalance in the distribution of data between STD and other classes has yet to be resolved. The following subsection introduces the N-best outputs scheme to enhance the recognition rate further and elevate the STD recognition rate to over 90%. This scheme is detailed in the subsequent section 6.3.1.

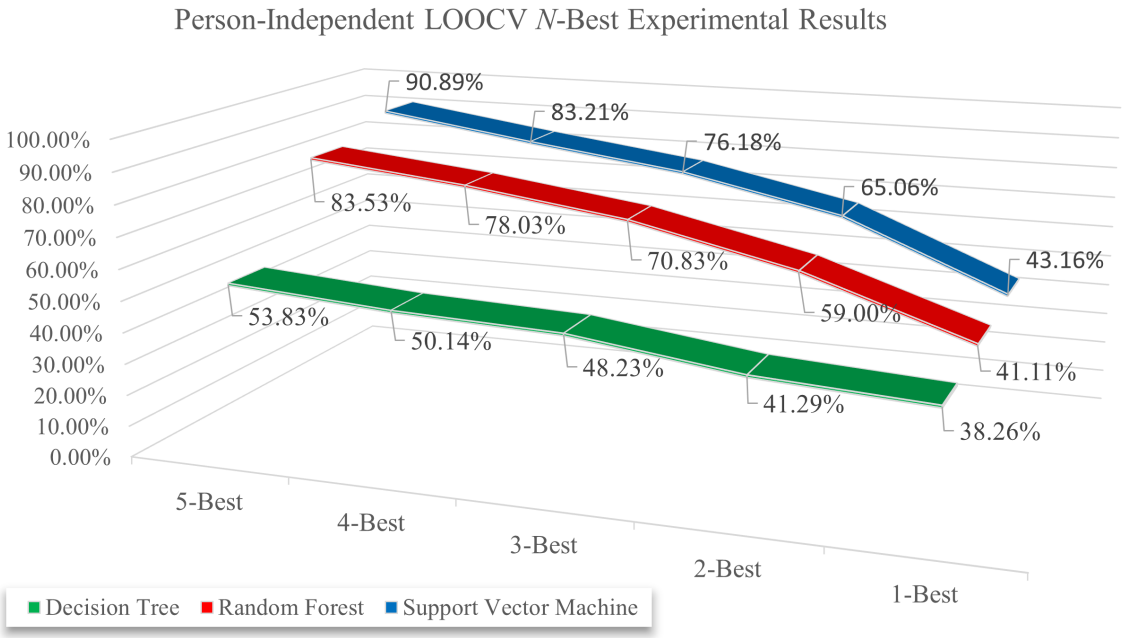

Figure 6.9: Recognition rates of one-tier person-independent N-best leave-one-out cross-validation (LOOCV) experiments with three non-deep machine learning models applying 11 statistical domain features. Each accuracy is the top result of the selection of the joint window length and overlap ratio[LZG+24].

N-Best outputs

The N-best output mechanism aims to present users with N recognition results having the highest probabilities, providing them with various scenarios to explore while enhancing the recognition accuracy of the person-independent model. Relying solely on a single output places considerable demands on the model's recognition accuracy, as an incorrect result may disrupt proper signal collection. Conversely, offering N potential results allows for greater tolerance of correct recognition rates and will enable users to conduct a more thorough examination. Metrics such as precision, recall, F1-score, and accuracy for the N-best outputs are calculated assuming that if only one of the N outputs corresponds to the labeled class, then the recognition is deemed correct. Consequently, an increase in N leads to a notable improvement in the precision of model recognition.

The figure 6.9 displays the offline recognition rates of the three models from 1 to 5 best outputs in person-independent experiments. Each experiment determines the best recognition rate through leave-one-out cross-validation (LOOCV), considering various combinations of window length and overlap ratio. As shown, there is a clear trend of improvement in recognition rate as N increases. Notably, the figure 6.10 presents that the 5-best recognition rate of SVM has reached 90.89%.

6.3.2 Results and discussion

Examining the results depicted in the figure 6.10, certain artifact types, notably TA-I, TA-E-O, TA-D-P, TA-N, and TA-D-G, demand special attention as they represent the primary sources of identification errors, with TA-M and TA-S being

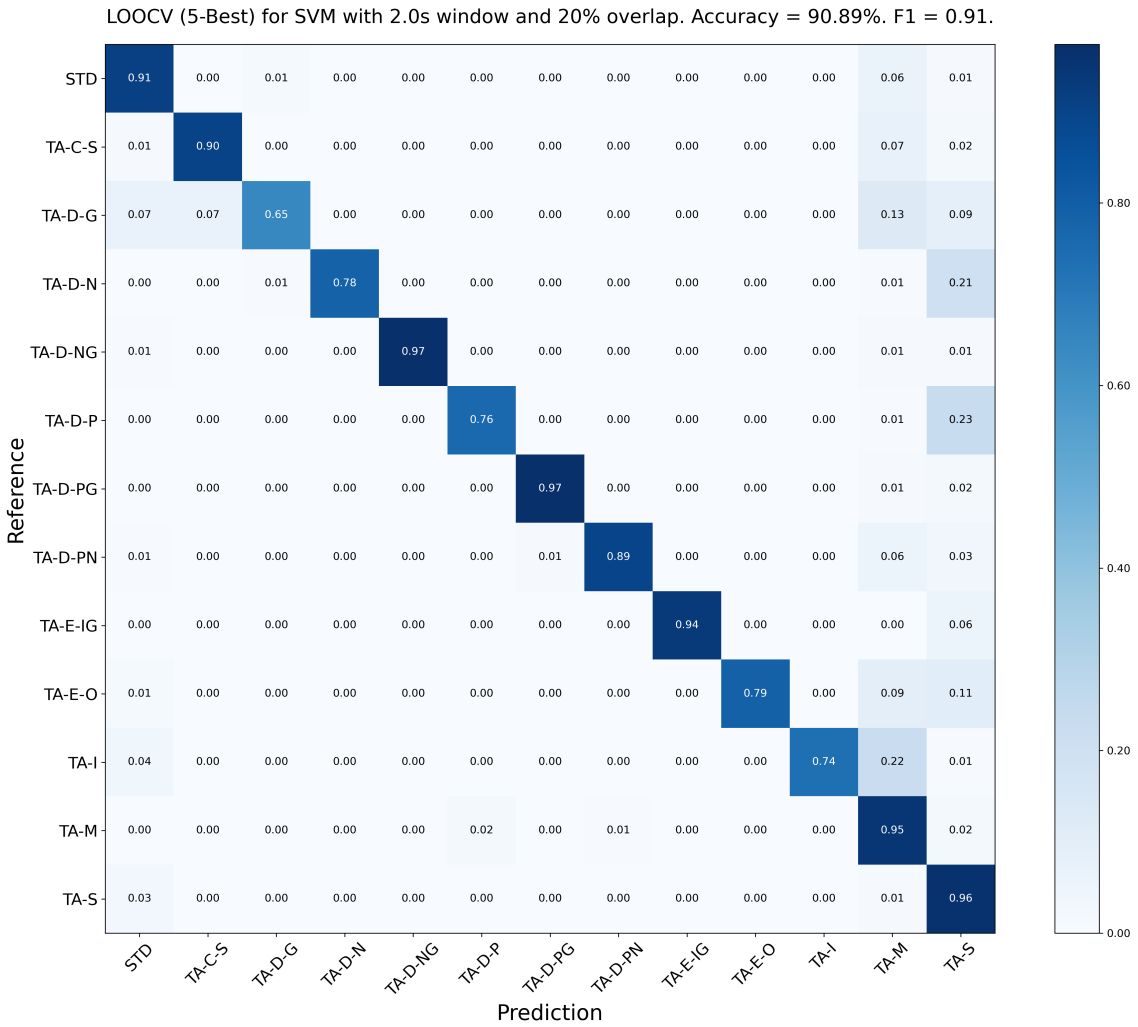

Figure 6.10: Confusion matrix of the one-tier support vector machine (SVM) modeling with 11 statistical domain features, 2-second window length, 20% overlap ratio, person-independent leave-one-out cross-validation (LOOCV), and 5-best outputs[LZG+24].

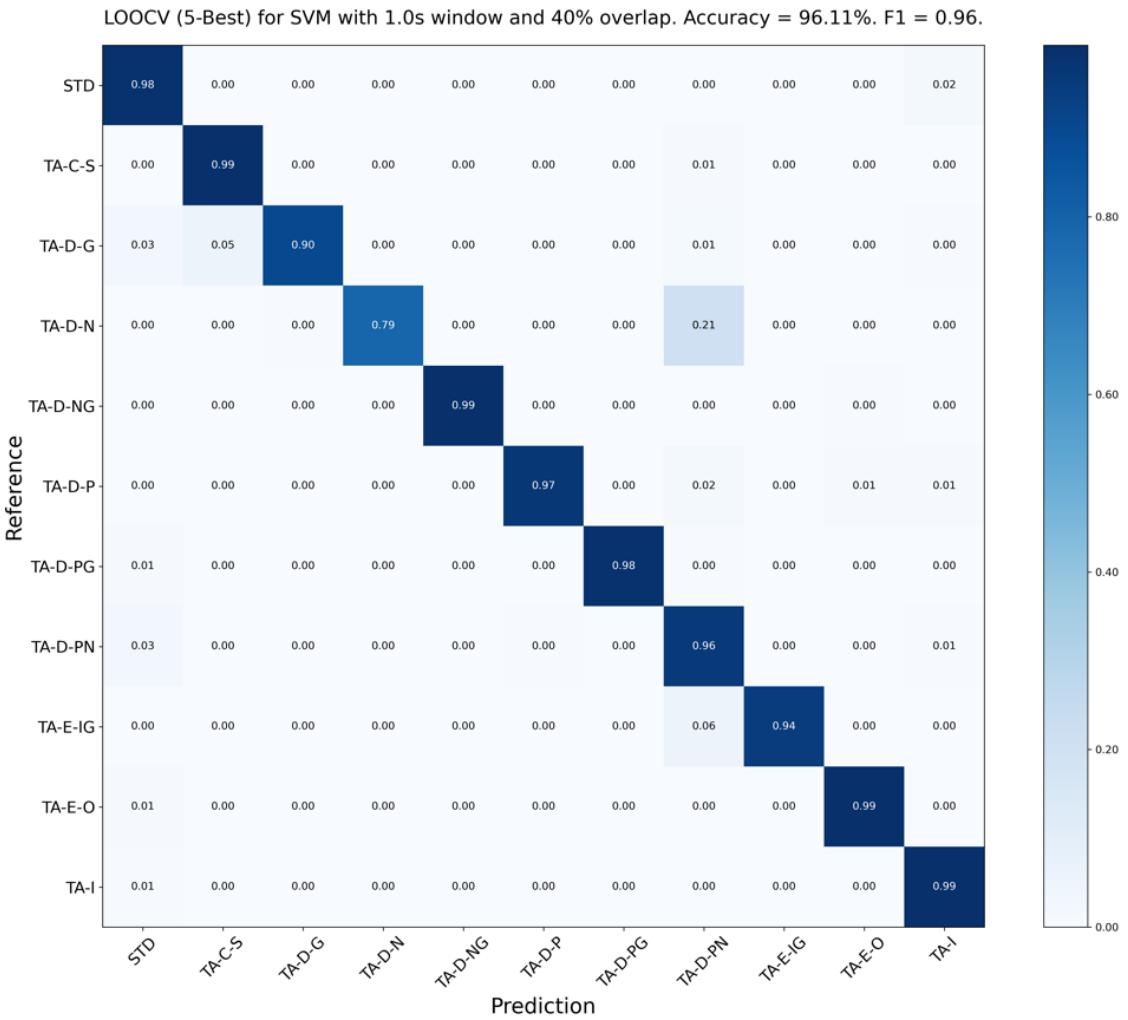

Figure 6.11: Confusion matrix of the support vector machine (SVM) modeling excluding TA-M and TA-S, with 11 statistical domain features, 1-second window length, 40% overlap ratio, person-independent leave-one-out cross-validation (LOOCV), and 5-best outputs[LZG+24].

the most commonly misattributed categories. TA-M and TA-S encompass various simulated events capable of triggering artifact generation. TA-M, in particular, involves different electrodes positioned near muscles generating force, resulting in a broader spectrum of ECG patterns covered within these two categories. Consequently, identifying representative features during machine learning becomes more challenging, significantly impacting model stability. TA-M and TA-S were temporarily excluded from the training dataset to validate this hypothesis, substantially improving model performance. As illustrated in the figure 6.11, the recognition accuracy of the support vector machine in the 5-bests PI LOOCV experiment increased to 96.11%. This discovery inspires further exploration into the separate identification of TA-M and TA-S. Additionally, it underscores the feasibility of the multilayer model proposed in this paper, wherein TA-S and TA-M could potentially be assigned distinct priorities—a topic warranting in-depth investigation in future studies.

Based on the analysis provided in Table 6.7, it is evident that semi-PI offers limited improvement potential in terms of performance. While it may lead to marginal enhancements in recognition rates for some N-best scenarios with RF and SVM,

Table 6.7: Recognition rates of N-best leave-one-out cross-validation experiments with the random forest (RF) and the support vector machine (SVM) modeling for comparing the performance of one-tier pure person-independent (PI) and semi-person-independent (semi-PI) training with 11 statistical domain features. semi-PI values inferior to PI are indicated in gray[LZG+24].

#-best	**RF:** PI	semi-PI	**SVM:** PI	semi-PI
1-best	41.11%	40.99%	43.16%	42.59%
2-best	59.00%	60.59%	65.06%	64.89%
3-best	70.83%	71.43%	76.18%	76.46%
4-best	78.03%	77.87%	83.21%	83.22%
5-best	83.53%	83.64%	90.89%	90.98%

there is no consistent trend of improvement. Moreover, due to the inherent dataset imbalance, employing semi-PI tends to exacerbate this issue. Although including STD from the test subjects resulted in a slight improvement in the STD recognition accuracy, it did not significantly alter the overall trend of recognition confusion. PI experiments without adding personal STD have achieved satisfactory identification accuracy in the case of n optimal outputs and using 2-M or H-M architectures.

6.4 From offline model towards real-time recognition

Parameters such as window length, overlap rate, features, and N size have been thoroughly evaluated in PI experiments. The best-performing and relatively fast parameter combinations are selected and saved to the model. For the preliminary experiments of real-time analysis, the relatively simple one-tier model architecture is retained and applied in real-time analysis. Visualization software for real-time signal acquisition is developed using the SDK provided by the acquisition equipment for signal collection. The software developed for real-time window-based artifact recognition segments the signal according to the window length and overlap ratio, inputs the final value into the trained and saved artifact classification and recognition model, and displays the N-Bests recognition results along with corresponding solutions. This allows users to conveniently check the collection environment and set up other interfering factors.

The figure 6.12 illustrates the interface of the visualization software. The upper section depicts real-time visualization for signal collection, while the lower section presents the N results with the highest prediction probability, classified for artifacts and arranged in order of probability. The prediction model was SVM, providing the highest accuracy among all the experiments. In the preliminary experiments, the results of the SVM model were first used as a reference. In the future development, the model will be comprehensively evaluated in combination with the recognition accuracy and operation speed, and the most suitable machine learning model for real-time analysis will be selected.

The comparison between window length and overlap ratio in the N-Best experiment reveals that high accuracy (90.89%) is achieved with a window length of 2 seconds and an overlap ratio of 20% as figure 6.13 presents the best-performing SVM model among all three ML models. However, a shorter step-size re-sults in a shorter delay of the recognition outcomes,but the interim recognition results may fluctuate dueto temporary search errors. On the other hand, longerdelay due to long step-sizes contradicts the character-istics of a real-time system, though it generates moreaccurate interim recognition results [LS19]. In preliminary experiments, a window length of 8 seconds was utilized, ensuring real-time performance at the expense of a slight decrease in accuracy.

6.4.1 Discussion

The experimental results of the real-time model demonstrate that ECG signals can be captured and visualized in real-time. Moreover, most artifacts can be detected, and recognition results are generated based on the recognition probability and sent to the user. However, the real-time model encounters several challenges in the initial experimental stage:

- The test environment lacked complete interference-free conditions. The presence of a power source resulted in the software consistently identifying TA-I (magnetic field interference), even during the standard ECG collection process. Dealing with power line interference is challenging in practice, as many collection devices require power, and computers—used as the application platform

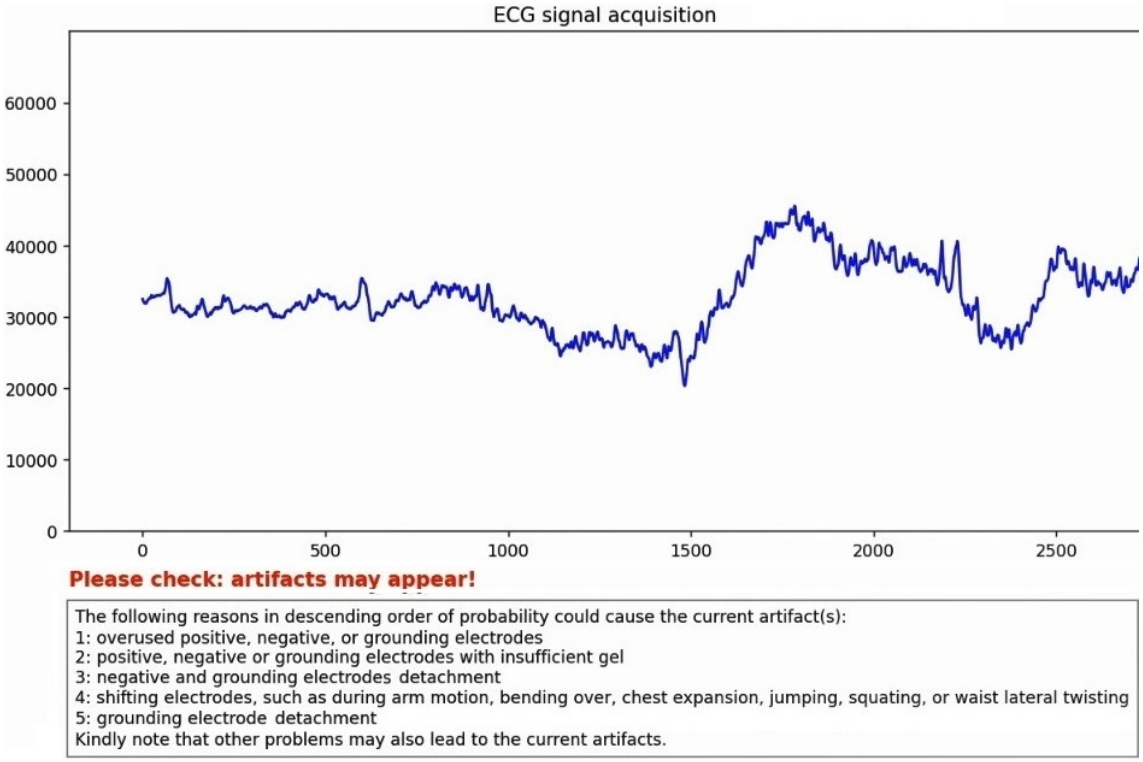

Figure 6.12: Screenshot of a running demonstration for real-time artifact detection and classification during ECG acquisition: the experimenter is being warned. Model: support vector machine (SVM) with one-tier person-independent training, 8-second window length and no overlap, 11 statistical domain features, and 5-best outputs[LZG+24].

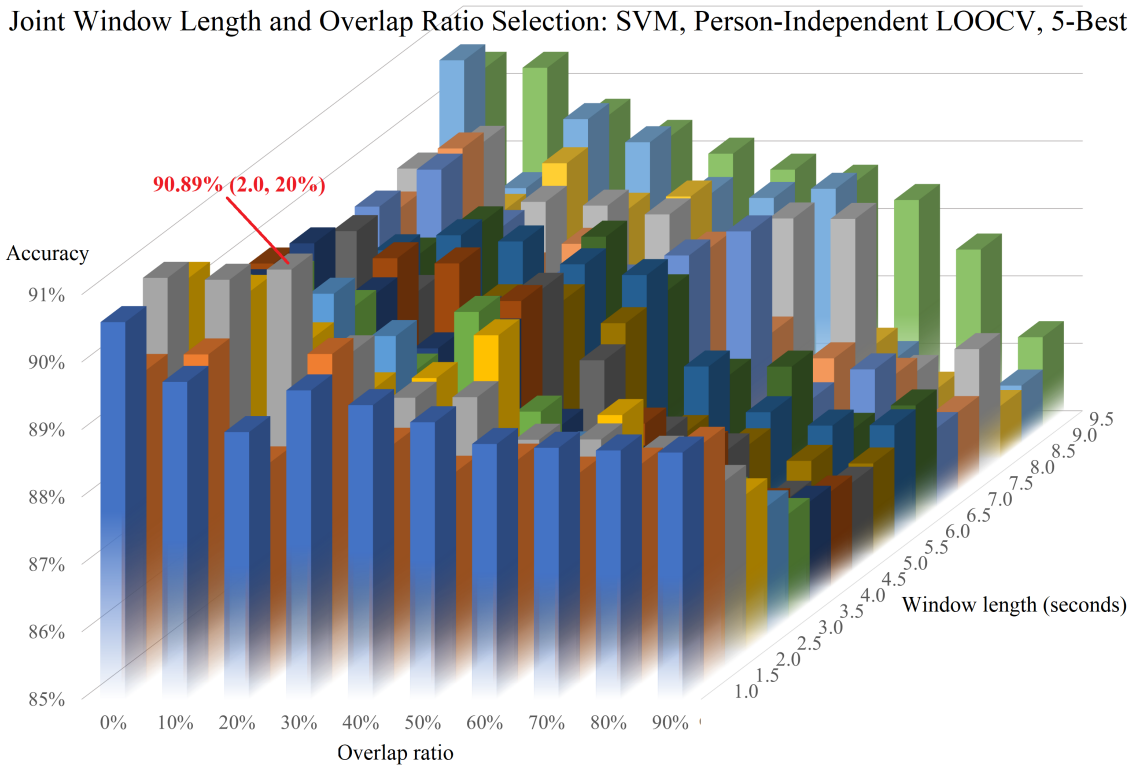

Figure 6.13: Instance of the joint window length and overlap ratio selection using heatmap: support vector machine (SVM) with 11 statistical domain features, person-independent leave-one-out cross-validation (LOOCV), one-tier modeling, and 5-best outputs[LZG+24].

for the software—need power to operate. However, signal preprocessing filters can filter out stable single artifacts like these. In future work, reducing the emphasis on power line interference and focusing solely on identifying significant electromagnetic interference could be considered.

- In real-time applications, a window length of 8 seconds and 0% overlap ratio were utilized. Longer window lengths were chosen over shorter ones due to their practicality. Short window lengths caused prompt information to refresh too rapidly, making it inconvenient for users to readjust. In contrast, longer window lengths effectively addressed this issue. However, offline model training favored a window length of 2 seconds for optimal performance. Thus, in practical scenarios, considerations extend beyond model performance alone, with user experience playing a crucial role. Balancing this trade-off and considering other factors requires further exploration in future work through additional experiments and user feedback. Additionally, a high overlap ratio ensures completeness of information in the signal but may introduce delays in real-time applications. This delay can desynchronize the real-time signal collection window and the cue information received by the user. Although the experiment employed a 0% overlap ratio, delay issues persisted during testing after prolonged use. This delay may occur during the model's recognition of signal fragments, which inherently requires event operations. Addressing the delay presents another challenge when transitioning from an offline model to an online analysis.

- The current real-time model architecture operates on a single-tier structure, enabling simultaneous recognition of all types. While this setup offers simplicity and ease of use, there's still room for performance enhancement. Despite the 5-best output's improvement, accurately identifying std remains pivotal for real-time artifact recognition. Future efforts will explore implementing a two-layer or multi-layer architecture to improve std recognition accuracy. Subsequently, the n-best output will be leveraged to enhance artifact classification accuracy accordingly.

- At present, SVM stands out for its superior recognition accuracy and serves as the artifact recognition model in real-time applications. However, it's crucial to acknowledge SVM's computational demands, especially as feature sets expand in the future. Furthermore, SVM's susceptibility to overfitting poses a challenge, potentially undermining its ability to maintain stable performance during real-time recognition, despite its promising training results. Although random forest may not consistently outperform SVM in terms of accuracy across all experiments, it demonstrates superior stability in processing results compared to SVM in real-time applications, given its speed and model robustness.

6.5 Pattern-based classification

In previous experiments, it was observed that ECG signals with simulated artifacts often contain more than one type of artifact. For instance, powerline interference is common in ECG signals due to the acquisition environment, and motion artifacts may also include baseline wandering, high-frequency noise, EMG signal noise, and others. This multifaceted nature of artifacts may have contributed to the suboptimal performance of the model in the 1-Best experiment. To investigate whether enhancing the clarity of signal patterns can improve the model's performance, a new type of annotation, namely pattern-based annotation, is proposed. The subsection 6.5.1 provides a detailed explanation of this new approach. Building on the new annotation, PI experiments were conducted, and the detailed results are presented in the following subsection 6.5.3.

6.5.1 Annotation

The new annotation approach involves observing the signal in both time and frequency domains to roughly identify the start and end moments of signal segments containing artifacts. Subsequently, artifact clips are extracted based on these markers and added to the dataset with corresponding labels. The recognition accuracy of artifacts based on pattern calibration is higher than that of artifacts based on event calibration. However, this study only conducted preliminary annotations. For instance, there is no standardized annotation criterion for determining the start and end moments of artifacts. Additionally, only one person annotated the dataset, and the annotation necessitates an annotator with a solid understanding of artifact patterns who can identify them visually. Annotating the signals across the entire database is a highly time-consuming task, and finding a suitable annotator poses a significant challenge.

The new annotation categories shown in table 6.8, encompass artifact types.

Table 6.8: Taxonomy of real-time artifacts during ECG acquisition: pattern-based

pattern-based Annotation	
Abbreviation	Explanation of ECG Artifact
Baseline wander	Usually caused by deep respiration or chest/body movement
Corrupted ECG	ECG beyound repair
EMG Noise	Electrical activity caused by muscle contractions
EMI Artifacts	reflection and adsorption of electromagnetic radiation
High Frequency Noise	Noise could be generated by other equipment
Inverted Electrodes	switched electrodes
Lead off	**E**lectrodes: left or right or both leads detachment
Motion Artifacts	the movement of the electrode away from its contact skin
No Grounding	Grounding lead detachment
Overused, Insufficient Gel	Insufficient gel causes loss of contact
Powerline Interference	Interference caused by power supply (50/60Hz)

6.5.2 Feature selection and training

To maintain controlled variables and enable a meaningful comparison of experimental outcomes, the features extracted from the newly annotated signals were deliberately kept consistent with those employed in both PD and PI experiments. Following the extraction of 11 statistical features, experiments were systematically carried out using Leave-One-Out Cross-Validation (LOOCV) on DT, SVM, and RF models. These experiments involved exploring different window lengths and overlap ratio settings, with the overarching goal of pinpointing reliable parameter combinations that optimize the overall performance of the models.

The experiment maintained the same division into two groups as in the PI setup. In one group, the aim was to classify all categories, forming a classification model with one-tire. This group encompassed various artifact types, STD, and corrupted ECGs. The other group initially concentrated on classifying only STD and corrupted ECGs. Subsequently, it expanded to classify data beyond STD and beyond STD and corrupted ECGs. These experiments were also designed to artificially reduce data variability to enhance recognition accuracy.

6.5.3 Results and discussion

After labeling based on patterns, the categories exhibit observable differences to the naked eye, which should facilitate machine learning classification compared to labels based on events. The table 6.9 presents a comparison of accuracy. The first and second columns display the accuracy of overall categorization in event-based versus pattern-based labeling, while the third and fourth columns depict the accuracy of standard ECGs in event-based versus pattern-based labeling.

The first row represents binary classification results, and rows two through six showcase overall classification outcomes. For comparison, the training results of the random forest are considered here. In terms of pattern-based classification, the performance of the RF is comparable to that of SVM. Several observations can be made:

- As hypothesized, pattern-based recognition yields significantly higher accuracy than event-based recognition. Figure 6.15 shows the confusion matrix of the 5-best pure PI LOOCV pattern-based with the highest overall recognition rate (87.07%), using RF modeling. The consideration for proposing pattern-based experiments stems from the fact that, in event-based signals, the same event may contain different artifact patterns. It is likely that the same pattern is present in different events, contributing to the poor performance of recognition results.

- Due to the new annotation, stricter theoretical requirements are imposed on standard ECGs. Consequently, the classification results for standard ECGs are notably superior to those of event-based annotation. This implies that artifact may be mixed in standard ECGs collected based on events, or other events may contain standard ECG fragments, resulting in poor recognition accuracy of standard ECGs.

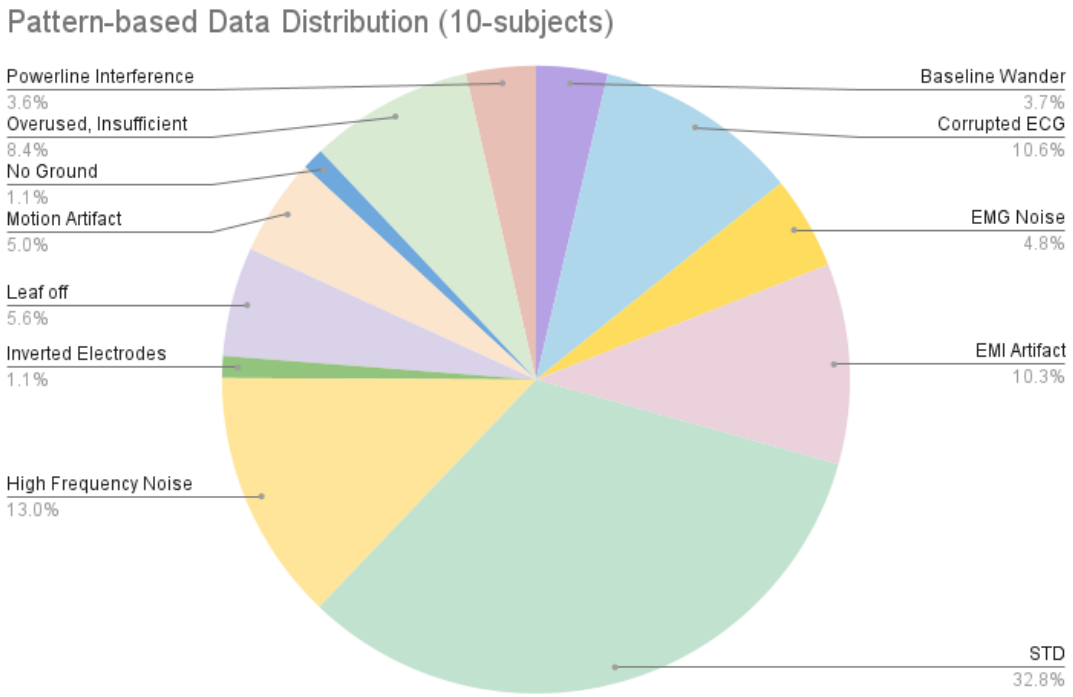

Figure 6.14: Data distribution: pattern-based annotation, 10-subject dataset.

- It is essential to consider the data distribution, which is shown in Figure 6.14. After pattern-based labeling, the proportion of STD in the data is even larger than in event-based labeled data, leading to a more pronounced imbalance in the database. This is evident in the full classification results in Figure 6.15, where overall recognition tends to favor STD. The displayed result in the figure 6.16 illustrates the removal of damaged ECG and STD signals, leading to an artifact recognition accuracy of 90.51%. Moreover, there is a notable improvement in the classification bias towards STD. This imbalance is a significant factor contributing to the substantial improvement in accuracy values of the classification results. Balancing the database in future work is crucial to instill higher confidence in the results.

- Among all classes annotated based on pattern, the classification of "No Grounding" yields the poorest results. The pattern of "No Grounding" closely resembles that of "STD" from an observational perspective, making it prone to confusion with "STD" in the overall classification. This confusion may stem from factors such as data imbalance and the lack of representative characteristics for the "No Grounding" class. Further research into the performance of this class is warranted for future work.

Table 6.9: Recognition rates of N-best leave-one-out cross-validation and binary classification between standard ECGs and corrupted ECGs experiments with the random forest (RF) modeling for comparing the performance of one-tier pure person-independent (PI) on event-based labels and pattern-based labels with 11 statistical domain features. events-based experiment values inferior to pattern-based experiments are indicated in gray.

Experiments	**RF:** event-based	pattern-based	**STD:** event-based	pattern-based
binary	82.01%	91.58%	53.00%	93.00%
1-best	41.11%	44.54%	40.00%	77.00%
2-best	59.00%	69.86%	66.00%	100.00%
3-best	70.83%	79.15%	83.00%	100.00%
4-best	78.03%	84.74%	84.00%	100.00%
5-best	83.53%	90.31%	91.00%	100.00%

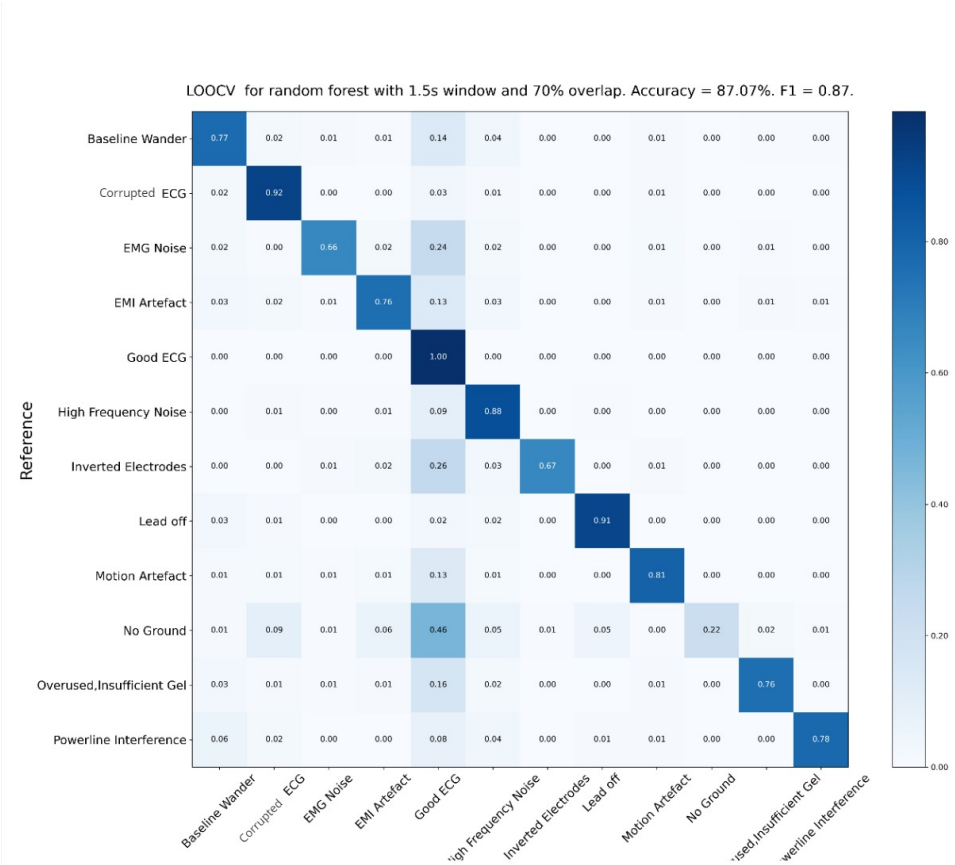

Figure 6.15: Confusion matrix of the one-tier random forest (RF) modeling with pattern-based annotation, 11 statistical domain features, 1.5-second window length,70% overlap ratio, person-independent leave-one-out cross-validation (LOOCV), and 5-best outputs.

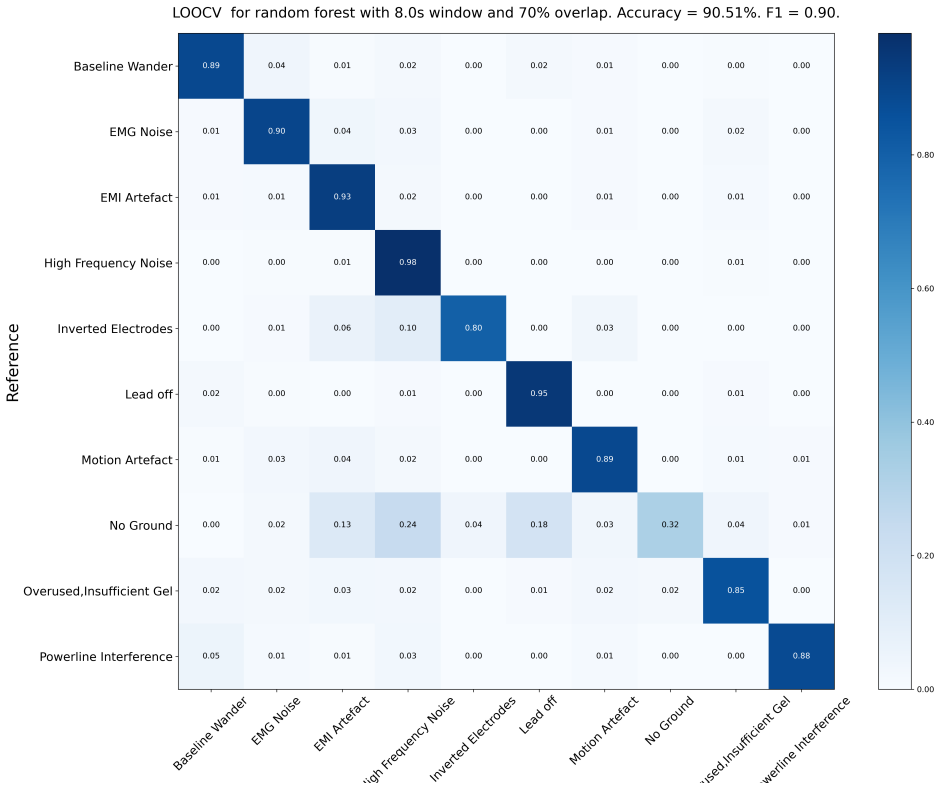

Figure 6.16: Confusion matrix of the one-tier random forest (RF) modeling with pattern-based annotation, removed STD and corrupted ECG classes, 11 statistical domain features, 8-second window length, 70% overlap ratio, person-independent leave-one-out cross-validation (LOOCV), and 5-best outputs.

7. Conclusion

Previous experiments have explored various artifact reduction and removal methods on collected ECG signals, including filters, wavelet transforms, and more. Additionally, real-time approaches for addressing artifacts include dichotomizing clean and noisy ECGs to prompt the user to recollect signals. Another method involves pattern-based identification of specific artifact types, such as baseline drift and EMG signal noise.

The limitation of these approaches is that artifact removal during post-processing may lead to the loss of vital signal information. Additionally, some artifacts resist removal by technological means, necessitating signal reacquisition. Real-time processing methods are confined to pattern-based recognition of artifacts, offering users insight into their types but lacking specific improvement suggestions. Binary classification results only indicate whether recollection is warranted without providing reasons for significant artifact presence in previously collected signals. These limitations persist, hampering the resolution of ECG collection challenges posed by artifacts.

This study promptly identifies artifacts during signal acquisition, and users are alerted upon detection. Users can adjust the acquisition device or environment to obtain high-quality ECG signals upon receiving the alert message.

The current real-time program verifies that the method proposed in this study can detect, classify, and alter artifacts in real-time during ecological acquisition. Experimental results with the offline model showed that the shorter the window and the higher the overlap rate, the better the model performance. However, a longer window of 8 seconds was chosen for the real-time demonstration to provide classification results, as it better suits the real-time processing needs. The overlap rate was set to 0% to minimize latency, as a higher overlap rate caused a delay, mainly when computational requirements were increased. Occasionally, artifacts caused by the environment (e.g., TA-I) could not be recognized in real-time during testing due to uncontrollable environmental parameters.

The offline model's performance could be improved by adding the tester's STD or providing N-Bests results. Individual variability in the characterization of artifacts is evident in person-independent experiments. A semi-PI setup, i.e., implementing

the recording of the tester's standard ECG signal, can help improve the recognition accuracy of STD. Still, significant improvement has yet to be made in the recognition of other artifacts. Applying N-Bests outputs enhances the accuracy of model recognition. Depending on the number of classes, N can be adjusted to provide more accurate notifications.

Pattern-based recognition improves the performance of the offline model, especially for standard ECGs. However, the calibration is too rough, lacks specific standards, and has an unbalanced data distribution, so the credibility of the results still needs to be considered. Meanwhile, although pattern-based recognition demonstrates the disadvantage of event-based recognition, i.e., the same triggering event contains more than consistently representative artifacts, it remains to be further investigated how pattern-based recognition can help the results of event-based recognition and how it can be applied in real-time analysis.

In conclusion, this study suggests the feasibility of treating artifacts as focal points of investigation. It proposes training machine learning models on databases containing various event-based artifacts and standard ECGs to apply these models for real-time artifact recognition during ECG acquisition. The single-layer architecture model showcased the best performance for the 5-best output, achieving 96% accuracy. For example, SVM with a 1-second window length, 40% overlap ratio, and 5-best PI LOOCV achieved recognition accuracy of 96.11% and an F1-score of 0.96.

8. Future work

This research provided a perspective that traditional ML methods also work for data with a predominantly error/noise/artifact/problem composition. After all, most research is based on (as much as possible) correct data [LZG+24]. Accurate analysis of the signal collection process and provision of relevant information to users are essential prerequisites for widespread adoption. Numerous aspects require further research and optimization in future work.

- The primary objective is to enhance the performance of offline models. This can be achieved through various strategies, such as optimizing features within existing machine learning models. Presently, only statistical domain features are employed. Features in the frequency domain can also play a crucial role in identifying ECG artifacts, such as electromagnetic interference and power line interference, which exhibit distinct frequencies. Moreover, optimizing model parameters holds the potential for further enhancing model performance. Deep learning offers another avenue for exploration in future research endeavors. However, given the current database's relatively small size, either expanding the database or exploring transfer learning may be taken into consideration in future work. In transfer learning, a model trained on an existing large dataset can serve as a feature extractor or be fine-tuned to classify artifacts using one's own training samples. While datasets like the MIT-BIH Arrhythmia [MM01] and PTB-XL ECG [WSB+20] datasets could be considered, it's important to note that most ECGs in large datasets originate from signals collected in hospitals and typically consist of 12 leads, unlike the 3 leads used in this study. Therefore, the adaptation of these large datasets would warrant further consideration in future research.

- Secondly, concerning the development of real-time software, the current experimental stage focuses on implementing essential functions. This includes real-time collection and visualization of ECG signals, segmenting the collected data based on window length and overlap ratio and extracting the features for model recognition. Moving forward, efforts will be made to refine the software by introducing additional features. Users will have the ability to

pause or terminate data collection upon artifact recognition, and they can decide whether to save, view, or discard collected signals afterward. These enhancements could be seamlessly integrated as add-ons to the signal collection software. Moreover, testing the real-time software should involve a broader user base, including individuals unfamiliar with specific development details. This approach will help assess whether the software design meets user needs and habits, facilitating further optimization.

- The study has demonstrated the feasibility of the proposed methodology and research architecture. ECG signals were chosen as the initial focus due to their single-channel nature, periodicity, consistent pattern, and simple structure. Moreover, their artifacts typically manifest in predictable forms. Looking ahead, this research methodology and process can be extended to include other signals such as EDA, RIP, and more, thereby expanding its applicability to various biological signals. Recognizing artifacts remains a formidable challenge across all biosignals types. Detecting them during the collection process can significantly enhance the subsequent analysis and research by improving the quality of biosignals.

A. Acronyms

- **ACC** Accelerometers
- **DT** Decision Tree
- **ECG** Electrocardiogram
- **EDA** Electrodermal activity
- **EEG** Electroencephalogram
- **EKG** Electrokardiogram
- **EMG** Electromyography
- **EMI** Electromagnetic interference
- **fNIRS** Functional near-infrared spectroscopy sensor
- **GYRO** Electrogoniometer
- **H-M** Hybrid multiple layers architecture
- **LASIS** Left anterior superior iliac spine
- **LC** Left clavicle, near left acromial end
- **LOOCV** Leave-one-out cross-validation
- **PD** Person-dependent
- **PI** Person-independent
- **PPG** Photoplethysmography
- **RIP** Respiratory inductance plethysmography

- **RC** Right clavicle, near right acromial end
- **RF** Random Forest
- **Semi-PI** Semi-person-independent
- **STD** Standard ECG waveforms
- **SVM** Support vector machine
- **TSFEL** Time-Series Feature Extraction Library
- **1-M** One-tier architecture
- **2-M** Two multiple layers architecture

B. Appendix

B.1 The 10-Subject dataset

Table B.1: 10-subjects dataset with all abbreviations of technical or biological artifacts, amount of segments, duration of each segment, and the total duration of each artifact.

Label	Segments	Duration/Segment (s/seg)	Total Duration
STD Standing	15	60	00:15:00
STD Sitting	14	60	00:14:00
BA-R-H	22	15	00:05:30
BA-R-R	22	15	00:05:30
BA-R-D	22	15	00:05:30
TA-S-C	10	30	00:05:00
TA-S-A	10	30	00:05:00
TA-S-S	10	30	00:05:00
TA-S-B	10	30	00:05:00
TA-S-J	10	30	00:05:00
TA-S-W	10	30	00:05:00
TA-D-P	10	30	00:05:00
TA-D-N	10	30	00:05:00
TA-D-G	10	30	00:05:00
TA-D-PG	10	30	00:05:00
TA-D-NG	10	30	00:05:00
TA-D-PN	10	30	00:05:00
TA-C-S	10	30	00:05:00
TA-M-P	10	30	00:05:00
TA-M-N	10	30	00:05:00
TA-M-G	10	30	00:05:00
TA-M-PN	10	30	00:05:00
TA-M-PNG	10	30	00:05:00
TA-M-PG	10	30	00:05:00
TA-M-NG	10	30	00:05:00
TA-E-IG	18	30	00:09:00
TA-I	9	30	00:04:30
TA-E-O	9	30	00:04:30

B.2 Experiments

B.2.1 Person-dependent experiment results

Table B.2: Recognition results of N-Best N-Folds cross-validation experiments with the random forest(RF), the support vector machine(SVM), and the decision tree(DT) of person-dependent training. All detailed parameters of each experiment are listed. "All" includes STD, all technical artifacts and biological artifacts. "T" refers to all classes of technical artifacts.

Label	Model	Win.	Over.	Folds	-Best	Features	Acc	F1
All	DT	8.5s	10%	4	1	8-stats	49.35%	0.53
STD+BAR,T	DT	8.5s	0%	4	1	8-stats	62.24%	0.63
STD,T	DT	7.5s	0%	4	1	8-stats	56.33%	0.59
T	DT	8.5s	0%	4	1	8-stats	63.54%	0.65
All	DT	6.0s	0%	4	1	11-stats	55.47%	0.56
STD+BAR,T	DT	7.5s	50%	4	1	11-stats	55.10%	0.56
STD,T	DT	6.5s	0%	4	1	11-stats	58.04%	0.59
T	DT	6.5s	0%	4	1	11-stats	55.77%	0.58
All	RF	9.5s	0%	4	1	8-stats	62.93%	0.63
STD+BAR,T	RF	2.5s	10%	4	1	8-stats	66.00%	0.69
STD,T	RF	8.0s	0%	4	1	8-stats	78.85%	0.80
T	RF	5.5s	10%	4	1	8-stats	66.67%	0.71
All	RF	9.0s	20%	4	1	11-stats	56.45%	0.59
STD+BAR,T	RF	7.0s	20%	4	1	11-stats	55.73%	0.53
STD,T	RF	9.0s	60%	4	1	11-stats	61.57%	0.63
T	RF	6.5s	60%	4	1	11-stats	63.54%	0.67
All	SVM	9.5s	20%	4	1	8-stats	70.69%	0.73
STD+BAR,T	SVM	8.0s	50%	4	1	8-stats	66.84%	0.69
STD,T	SVM	9.5s	20%	4	1	8-stats	71.15%	0.73
T	SVM	9.5s	20%	4	1	8-stats	68.75%	0.71
All	SVM	9.0s	60%	4	1	11-stats	56.25%	0.59
STD+BAR,T	SVM	9.0s	60%	4	1	11-stats	65.20%	0.68
STD,T	SVM	9.0s	60%	4	1	11-stats	62.50%	0.66
T	SVM	9.0s	60%	4	1	11-stats	64.50%	0.68
BAR1,2,3,T	DT	5.0s	0%	3	1	8-stats	38.10%	0.29
BAR1,2,3,T	DT	9.0s	40%	3	1	11-stats	46.15%	0.42
BAR1,2,3,T	RF	9.5s	10%	3	1	8-stats	33.33%	0.26
BAR1,2,3,T	RF	9.5s	0%	3	1	11-stats	38.10%	0.28
BAR1,2,3,T	SVM	9.5s	50%	3	1	8-stats	40.48%	0.30
BAR1,2,3,T	SVM	2.0s	40%	3	1	11-stats	32.26%	0.29
BAR2,3,T	DT	9.5s	20%	4	1	8-stats	58.65%	0.56
BAR2,3,T	DT	5.5s	0%	4	1	11-stats	58.15%	0.54
BAR2,3,T	RF	9.0s	10%	4	1	8-stats	65.79%	0.65
BAR2,3,T	RF	9.0s	40%	4	1	11-stats	63.25%	0.59
BAR2,3,T	SVM	9.5s	20%	4	1	8-stats	71.32%	0.70
BAR2,3,T	SVM	9.0s	20%	4	1	11-stats	69.18%	0.66

Table B.3: Recognition results of N-Best N-Folds cross-validation experiments with the random forest(RF), the support vector machine(SVM), and the decision tree(DT) of person-dependent training. All detailed parameters of each experiment are listed. + refers to merging two classes. "All" includes STD, all technical artifacts and biological artifacts. "T" refers to all classes of technical artifacts.

Label	Model	Win.	Over.	Folds	-Best	Features	Acc	F1
S+B1B2,B3,T	DT	8.5s	10%	4	1	8-stats	53.23%	0.53
S+B1B2,B3,T	DT	7.5s	30%	4	1	11-stats	48.08%	0.51
S+B1B2,B3,T	RF	8.5s	10%	4	1	8-stats	64.82%	0.54
S+B1B2,B3,T	RF	5.5s	90%	4	1	11-stats	54.55%	0.55
S+B1B2,B3,T	SVM	9.5s	20%	4	1	8-stats	59.06%	0.59
S+B1B2,B3,T	SVM	9.0s	20%	4	1	11-stats	55.30%	0.57
TAM+TAG,T	DT	7.5s	20%	4	1	8-stats	54.63%	0.58
TAM+TAG,T	DT	9.0s	60%	4	1	11-stats	74.51%	0.72
TAM+TAG,T	RF	2.5s	10%	4	1	8-stats	57.00%	0.58
TAM+TAG,T	RF	6.5s	60%	4	1	11-stats	77.43%	0.75
TAM+TAG,T	SVM	7.0s	80%	4	1	8-stats	63.21%	0.65
TAM+TAG,T	SVM	9.0s	60%	4	1	11-stats	76.96%	0.76
TAM+TAG,T,S	DT	1.5s	90%	4	1	8-stats	49.30%	0.47
TAM+TAG,T,S	DT	9.0s	60%	4	1	11-stats	64.41%	0.58
TAM+TAG,T,S	RF	2.5s	10%	4	1	8-stats	56.53%	0.55
TAM+TAG,T,S	RF	3.0s	20%	4	1	11-stats	63.79%	0.58
TAM+TAG,T,S	SVM	8.0s	50%	4	1	8-stats	54.55%	0.53
TAM+TAG,T,S	SVM	9.0s	60%	4	1	11-stats	64.83%	0.61

B.2.2 Semi-person-independent experiments with 11 statistical features, LOOCV, without BAR

Table B.4: Binary and triple recognition results of N-Best N-Folds cross-validation experiments with the random forest(RF), the support vector machine(SVM), and the decision tree(DT) of person-dependent training. All detailed parameters of each experiment are listed. + refers to merging two classes. "S" is the abbreviation of "STD", "B" is the abbreviation of "BA-R" and "T" refers to all technical artifacts.

Label (Bi,Tri)	Model	Win.	Over.	Folds	Features	Acc	F1
BAR,T	DT	9.0s	20%	4	8-stats	96.77%	0.97
STD+BAR,T	DT	9.5s	70%	4	8-stats	100.00%	1.00
STD,BAR	DT	1.0s	10%	4	8-stats	25.67%	0.35
STD,BAR,T	DT	6.0s	20%	4	8-stats	80.00%	0.80
STD,BAR+T	DT	1.0s	60%	4	8-stats	63.01%	0.57
STD,T	DT	7.0s	40%	4	8-stats	92.02%	0.93
BAR,T	DT	6.0s	20%	4	11-stats	90.80%	0.93
STD+BAR,T	DT	9.5s	90%	4	11-stats	100.00%	1.00
STD,BAR	DT	1.5s	60%	4	11-stats	23.49%	0.24
STD,BAR,T	DT	8.0s	30%	4	11-stats	79.65%	0.82
STD,BAR+T	DT	1.0s	60%	4	11-stats	58.04%	0.64
STD,T	DT	7.0s	40%	4	11-stats	92.02%	0.93
BAR,T	RF	7.0s	30%	4	8-stats	99.34%	0.99
STD+BAR,T	RF	7.0s	80%	4	8-stats	100.00%	1.00
STD,BAR	RF	9.5s	10%	4	8-stats	33.33%	0.50
STD,BAR,T	RF	6.0s	20%	4	8-stats	83.22%	0.86
STD,BAR+T	RF	5.5s	0%	4	8-stats	78.54%	0.80
STD,T	RF	9.5s	10%	4	8-stats	81.09%	0.85
BAR,T	RF	1.0s	20%	4	11-stats	91.33%	0.91
STD+BAR,T	RF	9.5s	90%	4	11-stats	100.00%	1.00
STD,BAR	RF	1.5s	60%	4	11-stats	23.49%	0.24
STD,BAR,T	RF	9.5s	40%	4	11-stats	72.09%	0.76
STD,BAR+T	RF	1.0s	60%	4	11-stats	58.23%	0.64
STD,T	RF	7.0s	10%	4	11-stats	94.70%	0.95
BAR,T	SVM	9.5s	50%	4	8-stats	99.43%	0.99
STD+BAR,T	SVM	9.5s	90%	4	8-stats	100.00%	1.00
STD,BAR	SVM	9.5s	10%	4	8-stats	33.33%	0.50
STD,BAR,T	SVM	9.5s	10%	4	8-stats	84.36%	0.87
STD,BAR+T	SVM	9.5s	20%	4	8-stats	68.33%	0.73
STD,T	SVM	9.0s	70%	4	8-stats	100.00%	1.00
BAR,T	SVM	6.0s	20%	4	11-stats	90.80%	0.93
STD+BAR,T	SVM	9.5s	90%	4	11-stats	100.00%	1.00
STD,BAR	SVM	1.0s	20%	4	11-stats	24.28%	0.25
STD,BAR,T	SVM	7.0s	70%	4	11-stats	74.04%	0.77
STD,BAR+T	SVM	1.0s	60%	4	11-stats	57.86%	0.63
STD,T	SVM	7.0s	60%	4	11-stats	92.57%	0.93

Table B.5: Binary and triple recognition results of N-Best N-Folds cross-validation experiments with the random forest(RF), the support vector machine(SVM), and the decision tree(DT) of person-dependent training. All detailed parameters of each experiment are listed. + refers to merging two classes. "S" is the abbreviation of "STD", "B" is the abbreviation of "BA-R" and "T" refers to all technical artifacts.

Label (Bi,Tri)	Model	Win.	Over.	Folds	Features	Acc	F1
BAR1,2,3	DT	9.5s	20%	3	8-stats	77.78%	0.75
BAR1,2,3	DT	8.5s	10%	3	11-stats	88.89%	0.88
BAR1,2,3	RF	9.5s	50%	3	8-stats	66.67%	0.62
BAR1,2,3	RF	9.5s	0%	3	11-stats	88.89%	0.89
BAR1,2,3	SVM	9.5s	50%	3	8-stats	94.44%	0.94
BAR1,2,3	SVM	5.0s	0%	3	11-stats	94.44%	0.94
STDBAR1,BAR23	DT	1.5s	60%	3	8-stats	25.72%	0.22
STDBAR1,BAR23	DT	1.5s	60%	3	11-stats	25.72%	0.22
STDBAR1,BAR23	RF	1.5s	60%	3	11-stats	26.61%	0.23
STDBAR1,BAR23	SVM	9.5s	10%	3	8-stats	100.00%	1.00
STDBAR1,BAR23,T	DT	1.5s	60%	4	8-stats	11.21%	0.08
STDBAR1,BAR23,T	DT	1.5s	60%	4	11-stats	11.21%	0.07
STDBAR1,BAR23,T	RF	9.0s	20%	4	8-stats	90.28%	0.87
STDBAR1,BAR23,T	RF	8.5s	30%	4	11-stats	75.00%	0.70
STDBAR1,BAR23,T	SVM	7.5s	0%	4	8-stats	88.89%	0.84
STDBAR1,BAR23,T	SVM	8.5s	30%	4	11-stats	75.00%	0.69

Table B.6: Recognition results of N-Best leave-one-out cross-validation(LOOCV) experiments with the random forest(RF), the support vector machine(SVM), and the decision tree(DT) of semi-person-independent training. All detailed parameters of each experiment are listed. "All" includes STD, and all technical artifacts. "T" refers to all classes of technical artifacts.

Label	Model	Win.	Over.	Folds	-Best	Acc	F1
All	DT	3.0s	30%	10	1	38.26%	0.38
All	DT	8.5s	30%	10	2	41.29%	0.41
All	DT	8.5s	30%	10	3	48.23%	0.47
All	DT	8.5s	30%	10	4	50.14%	0.48
All	DT	6.5s	50%	10	5	53.83%	0.51
All	RF	1.0s	10%	10	1	40.99%	0.39
All	RF	4.5s	0%	10	2	60.59%	0.59
All	RF	1.0s	40%	10	3	71.43%	0.70
All	RF	1.0s	0%	10	4	77.87%	0.76
All	RF	1.0s	30%	10	5	83.64%	0.83
All	SVM	9.5s	0%	10	1	42.59%	0.37
All	SVM	1.0s	10%	10	2	64.89%	0.61
All	SVM	9.5s	10%	10	3	76.46%	0.75
All	SVM	1.0s	20%	10	4	83.22%	0.82
All	SVM	2.0s	20%	10	5	90.93%	0.91

B.2.3 Person-independent experiments with 11 statistical features, LOOCV, without BAR

Table B.7: Recognition results of N-Best leave-one-out cross-validation(LOOCV) experiments with the random forest(RF) and the decision tree(DT) of person-independent training. All detailed parameters of each experiment are listed. "All" includes STD, and all technical artifacts. "T" refers to all classes of technical artifacts.

Label	Model	Win.	Over.	Folds	-Best	Acc	F1
All	DT	8.0s	60%	7	2	50.23%	0.49
All	DT	8.5s	20%	7	3	56.09%	0.55
All	DT	8.5s	20%	7	4	58.85%	0.58
All	DT	8.5s	20%	7	5	60.18%	0.59
without TAM TAEO	DT	2.5s	70%	7	2	59.50%	0.57
without TAM TAEO	DT	8.0s	0%	7	3	59.97%	0.59
without TAM TAEO	DT	2.5s	70%	7	4	63.77%	0.62
without TAM TAEO	DT	3.0s	50%	7	5	70.20%	0.68
T	DT	6.0s	80%	7	2	53.84%	0.52
T	DT	6.0s	80%	7	3	60.21%	0.57
T	DT	8.5s	90%	7	4	71.58%	0.66
T	DT	8.0s	90%	7	5	74.73%	0.70
T without TAM TAEO	DT	7.5s	70%	7	2	63.55%	0.63
T without TAM TAEO	DT	8.0s	60%	7	3	69.16%	0.69
T without TAM TAEO	DT	2.0s	20%	7	4	73.19%	0.70
T without TAM TAEO	DT	3.5s	50%	7	5	76.53%	0.75
All	RF	8.0s	10%	7	2	62.64%	0.62
All	RF	8.0s	80%	7	3	72.81%	0.72
All	RF	7.0s	0%	7	4	77.89%	0.77
All	RF	8.5s	40%	7	5	81.75%	0.81
without TAM TAEO	RF	6.0s	0%	10	2	60.83%	0.59
without TAM TAEO	RF	6.5s	0%	10	3	69.72%	0.68
without TAM TAEO	RF	9.5s	0%	10	4	75.86%	0.74
without TAM TAEO	RF	9.5s	50%	10	5	83.08%	0.82
T	RF	5.5s	90%	10	2	59.85%	0.58
T	RF	7.0s	10%	10	3	69.35%	0.68
T	RF	8.5s	30%	10	4	77.50%	0.77
T	RF	5.5s	40%	10	5	82.53%	0.82
T without TAM TAEO	RF	1.0s	0%	10	2	63.86%	0.61
T without TAM TAEO	RF	6.0s	30%	10	3	74.17%	0.73
T without TAM TAEO	RF	7.0s	10%	10	4	79.06%	0.78
T without TAM TAEO	RF	7.5s	20%	10	5	82.59%	0.81

Table B.8: Recognition results of N-Best leave-one-out cross-validation(LOOCV) experiments with the support vector machine(SVM)of person-independent training. All detailed parameters of each experiment are listed. "All" includes STD, and all technical artifacts. "T" refers to all classes of technical artifacts.

Label	Model	Win.	Over.	Folds	-Best	Acc	F1
All	SVM	8.5s	40%	7	2	62.83%	0.61
All	SVM	9.0s	20%	7	3	72.98%	0.70
All	SVM	7.5s	80%	7	4	77.90%	0.75
All	SVM	5.5s	30%	7	5	82.44%	0.80
without TAM TAEO	SVM	4.5s	80%	7	2	71.34%	0.67
without TAM TAEO	SVM	5.0s	30%	7	3	78.29%	0.74
without TAM TAEO	SVM	4.5s	60%	7	4	82.27%	0.80
without TAM TAEO	SVM	5.5s	90%	7	5	85.75%	0.84
T	SVM	9.0s	30%	7	2	64.88%	0.60
T	SVM	9.0s	30%	7	3	65.84%	0.62
T	SVM	9.5s	10%	7	4	78.41%	0.75
T	SVM	5.5s	30%	7	5	80.84%	0.78
T without TAM TAEO	SVM	4.5s	30%	7	2	69.72%	0.65
T without TAM TAEO	SVM	3.5s	40%	7	3	74.03%	0.69
T without TAM TAEO	SVM	2.0s	0%	7	4	77.76%	0.74
T without TAM TAEO	SVM	5.5s	90%	7	5	82.72%	0.80

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